



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Cloud infrastructures and services

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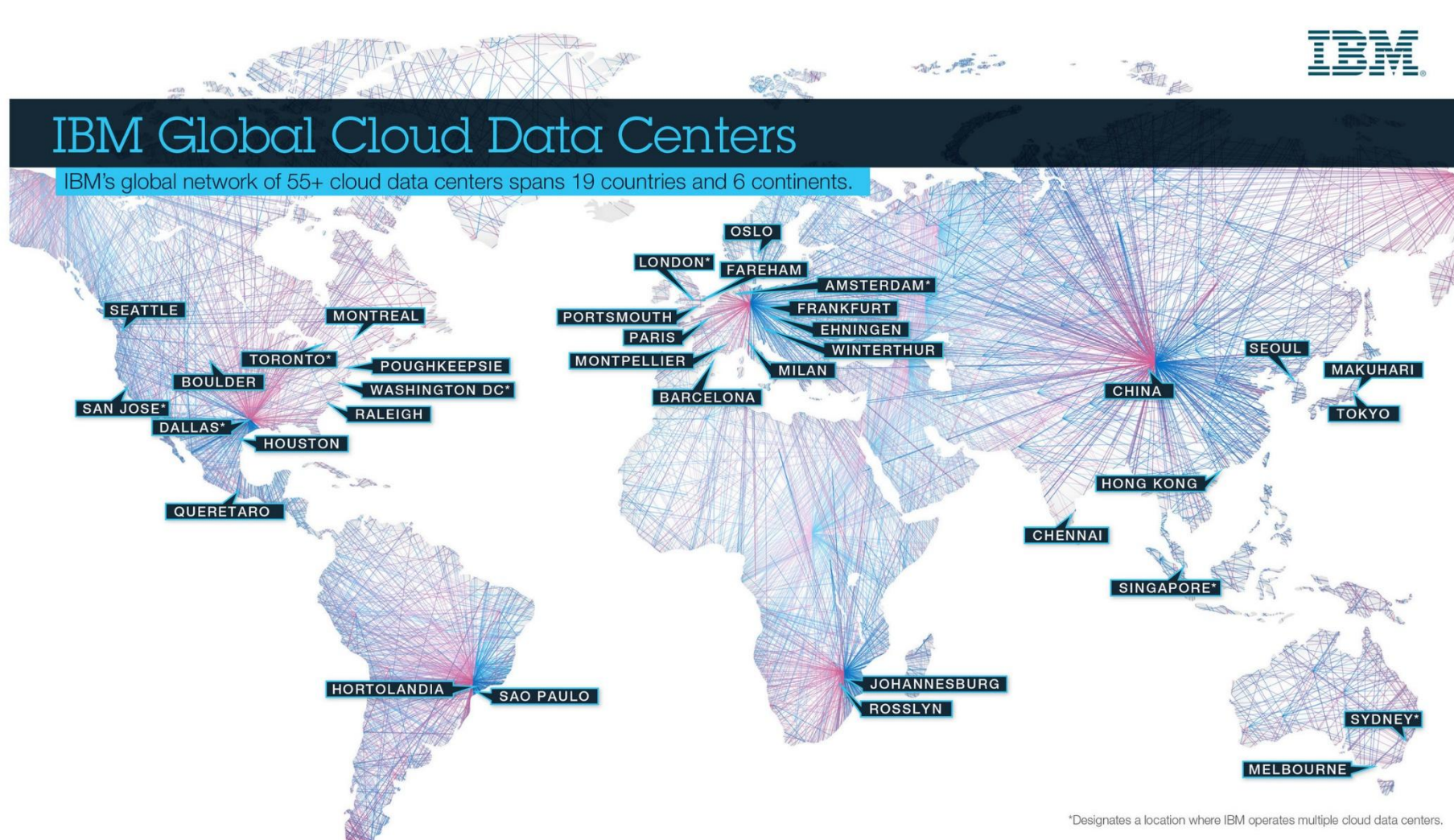
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Cloud infrastructures



Amazon Web Services





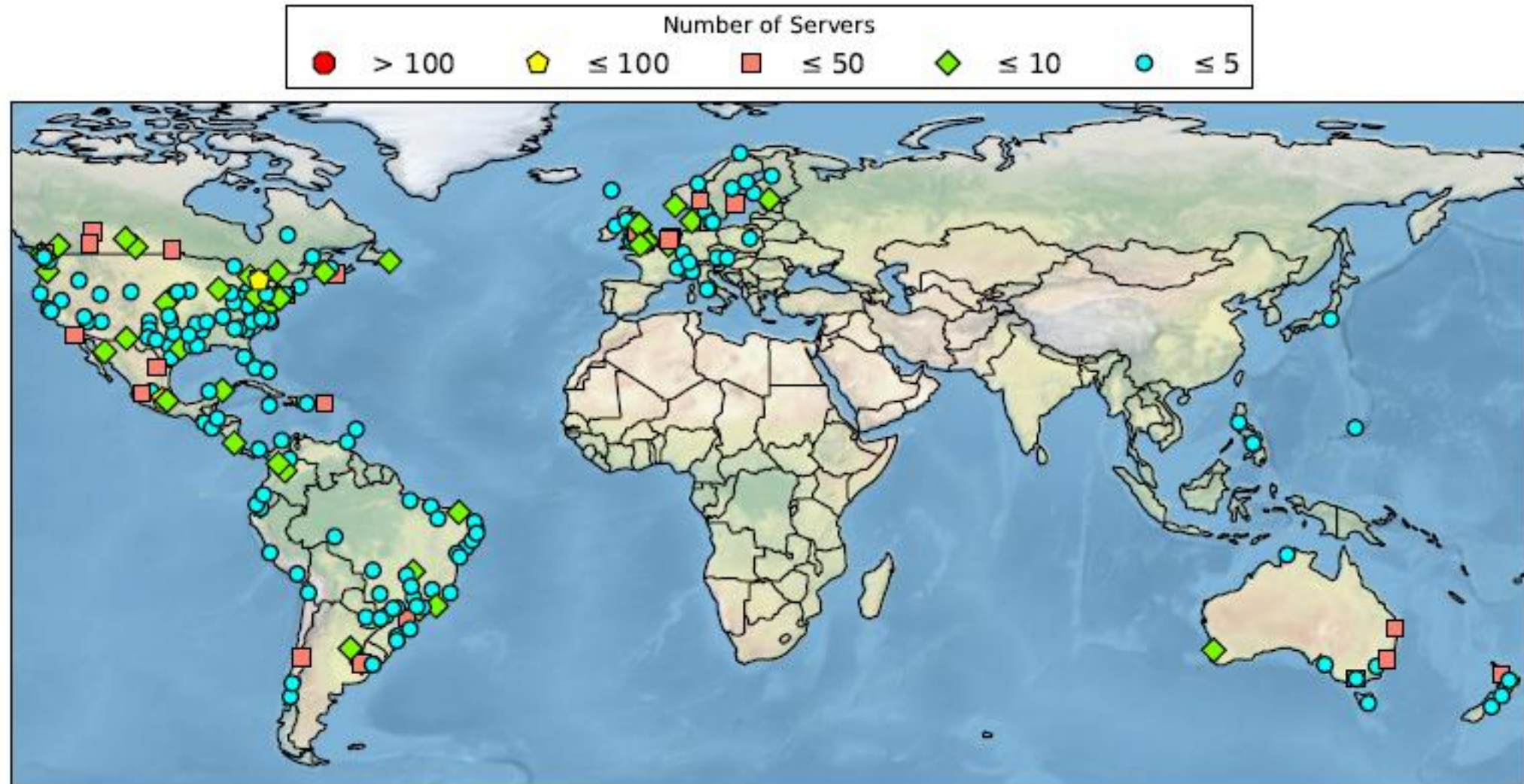
Datcenters of Salesforce



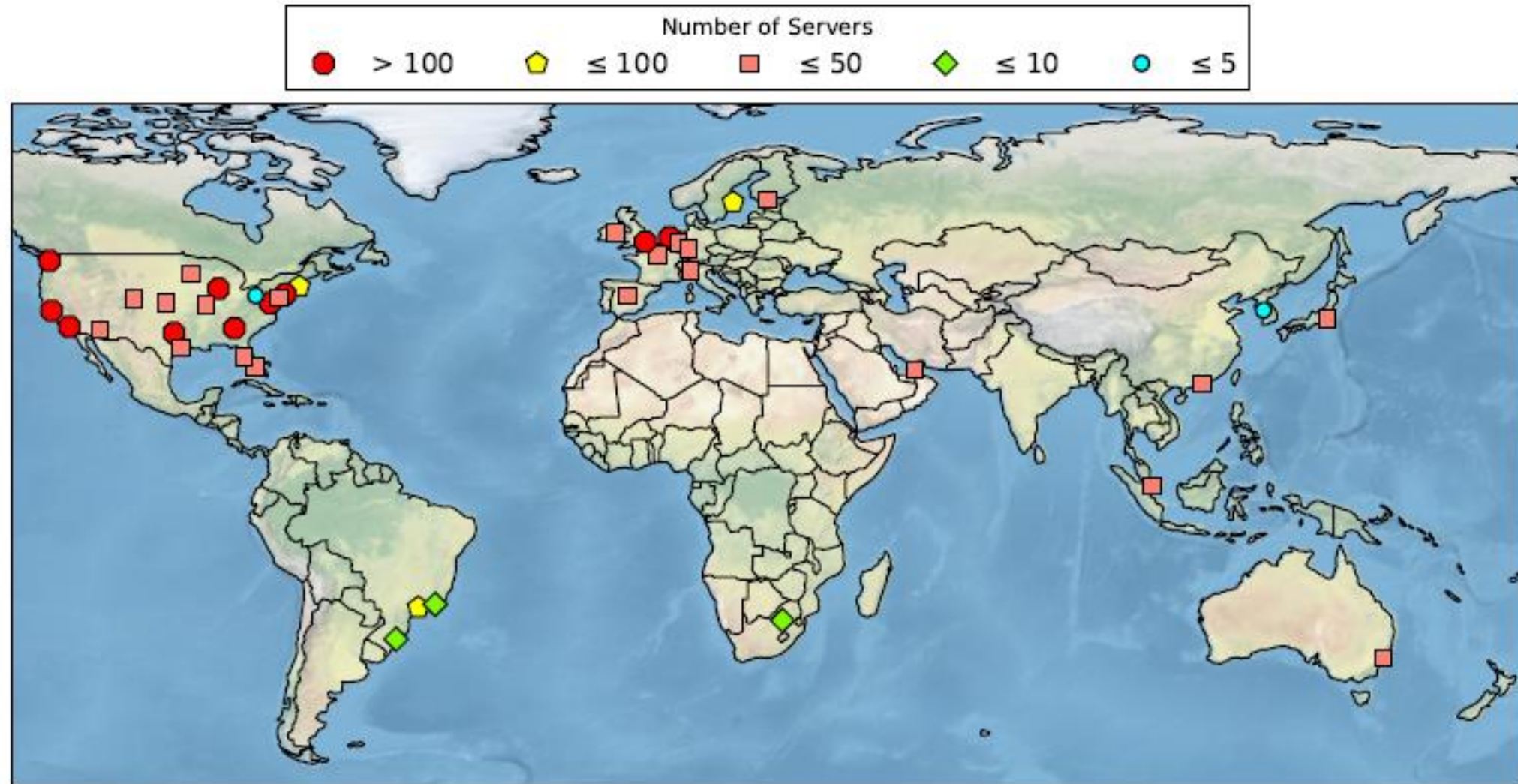
Microsoft Azure



Netflix servers deployed at ISP



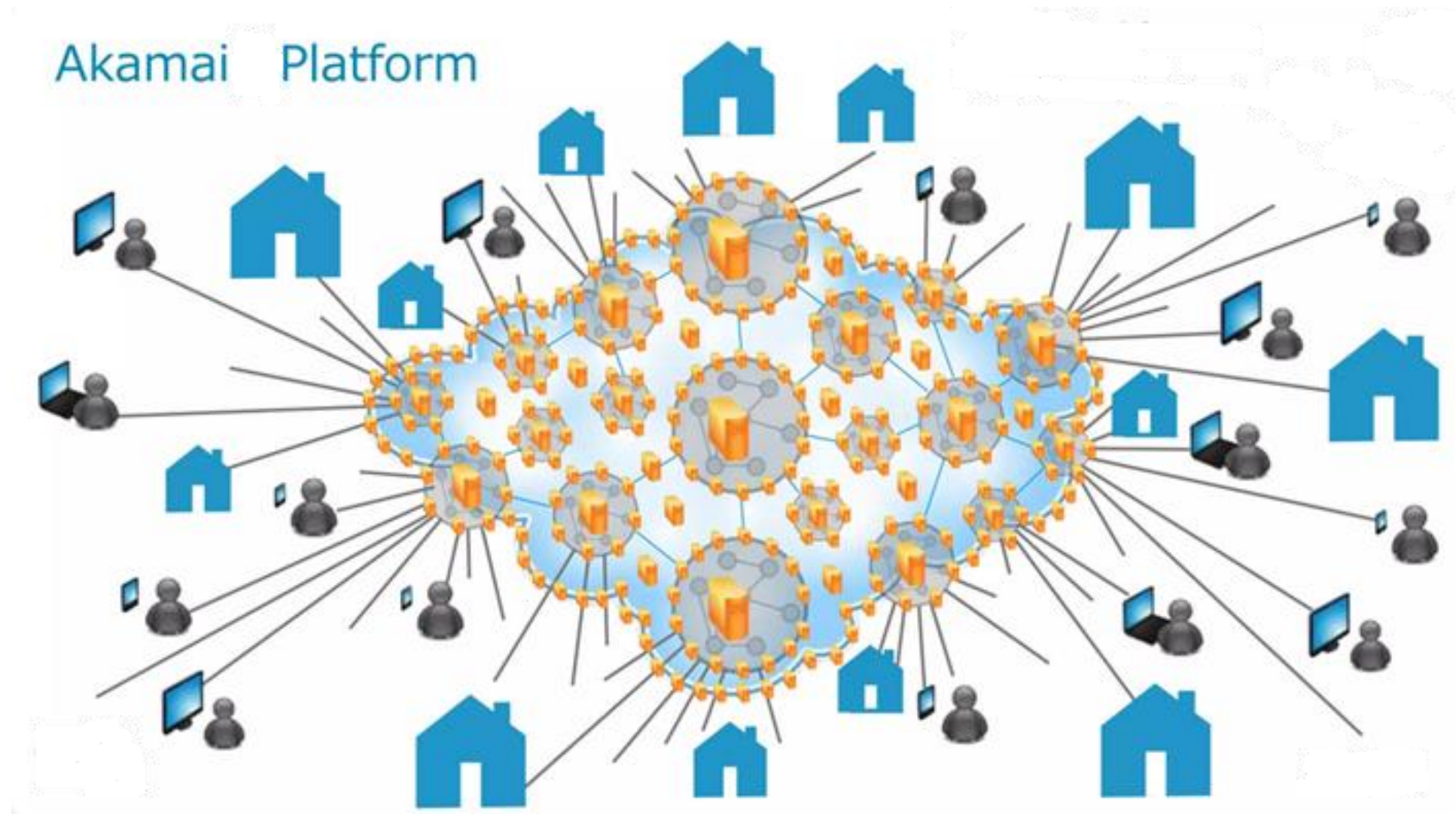
Netflix servers deployed at IXP



Dropbox edge servers



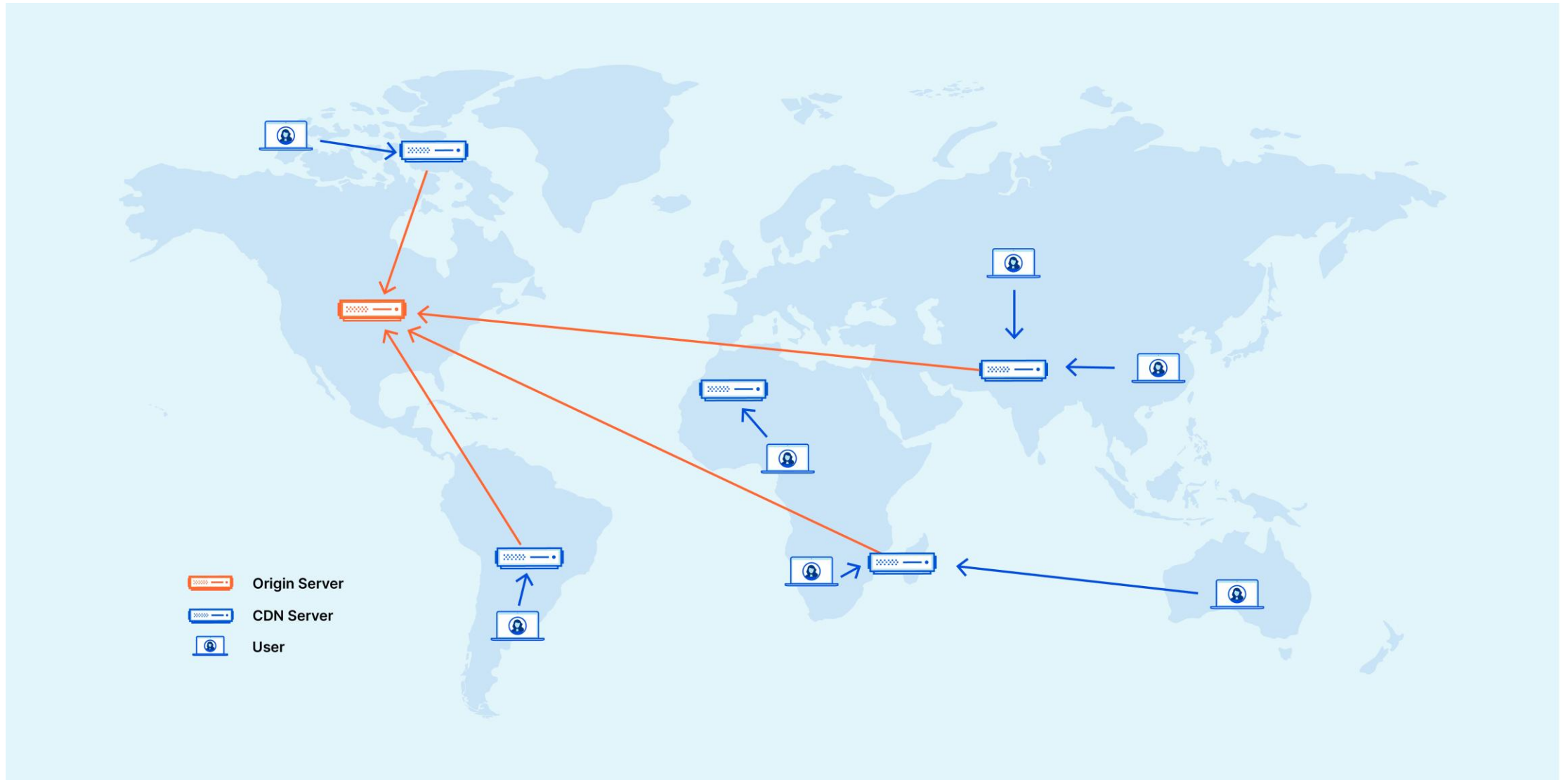
Akamai infrastructure



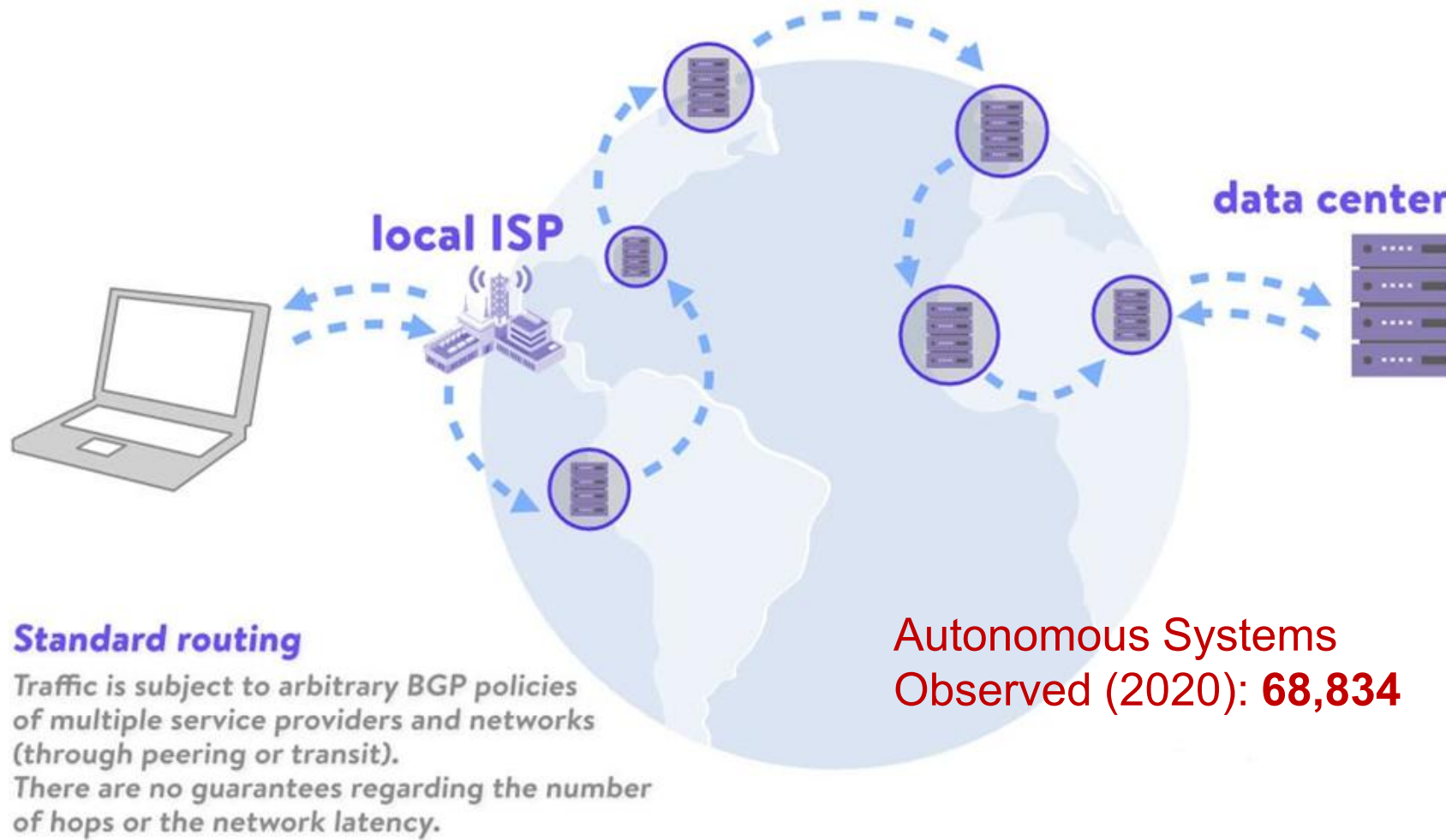
Cloudflare datacenters



CDN operations



The long path towards the “cloud”



The network impact on remote digital services

1. Last mile problem

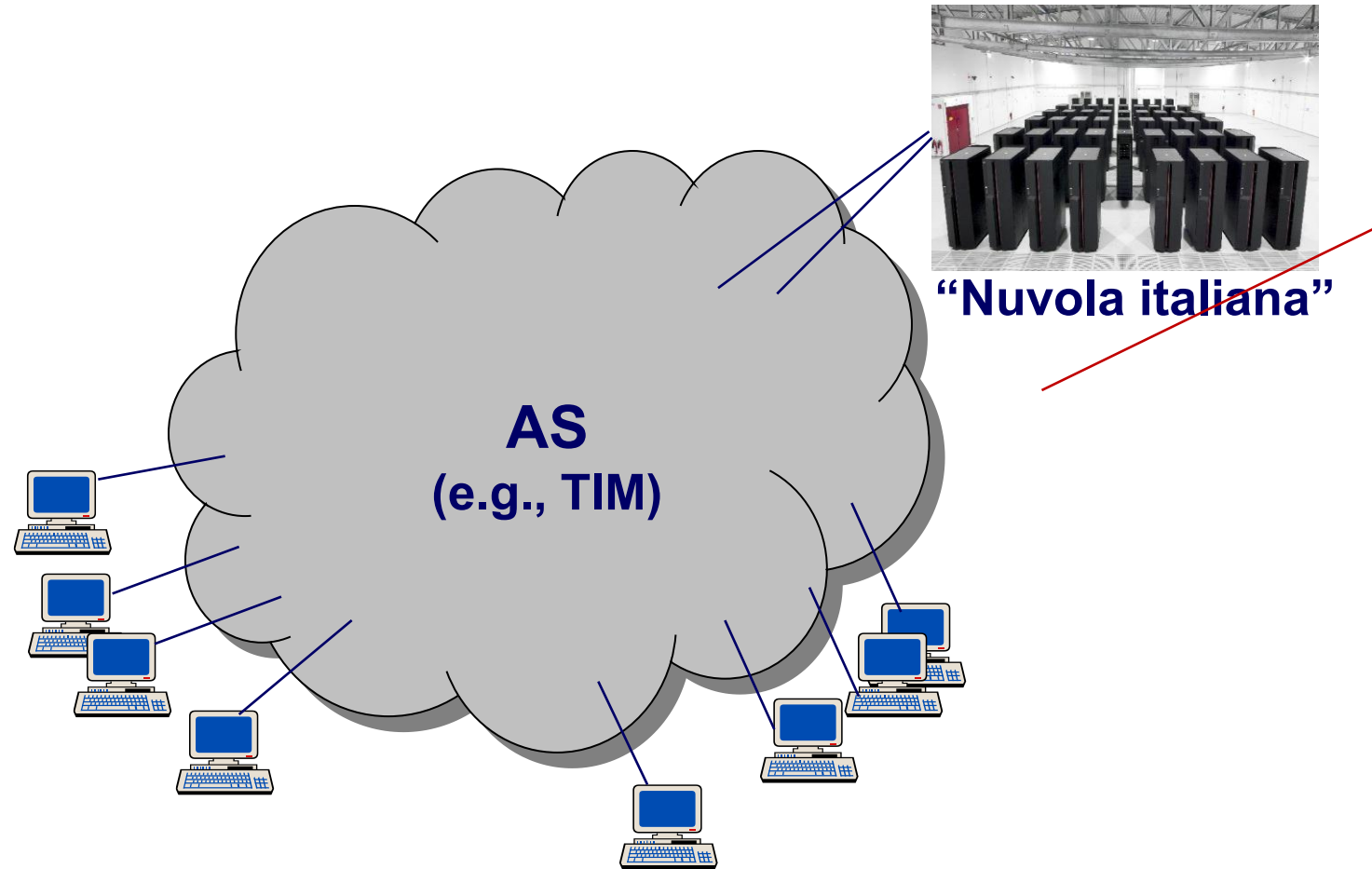
2. Network traffic

3. Congestion near the provider site (*first mile problem*)

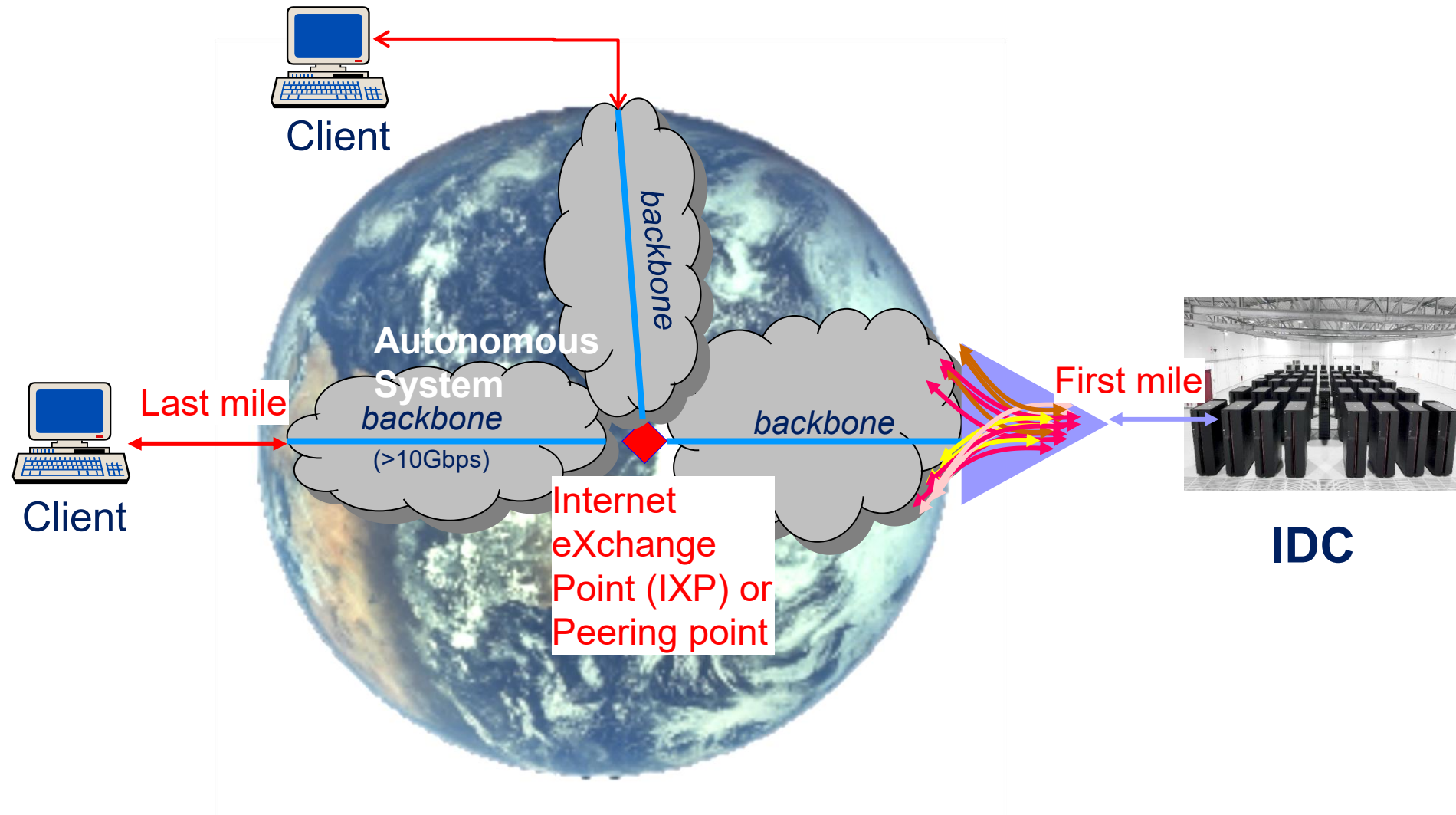
- As the number of users increases, services can become congested, like what happens to roads during rush hour. The scalability, instead of increasing, decreases with the number of users
- Cloud providers must take care to make sure that the congestion point is greater than the expected usage level by
 - increasing available capacity
 - replicating service points



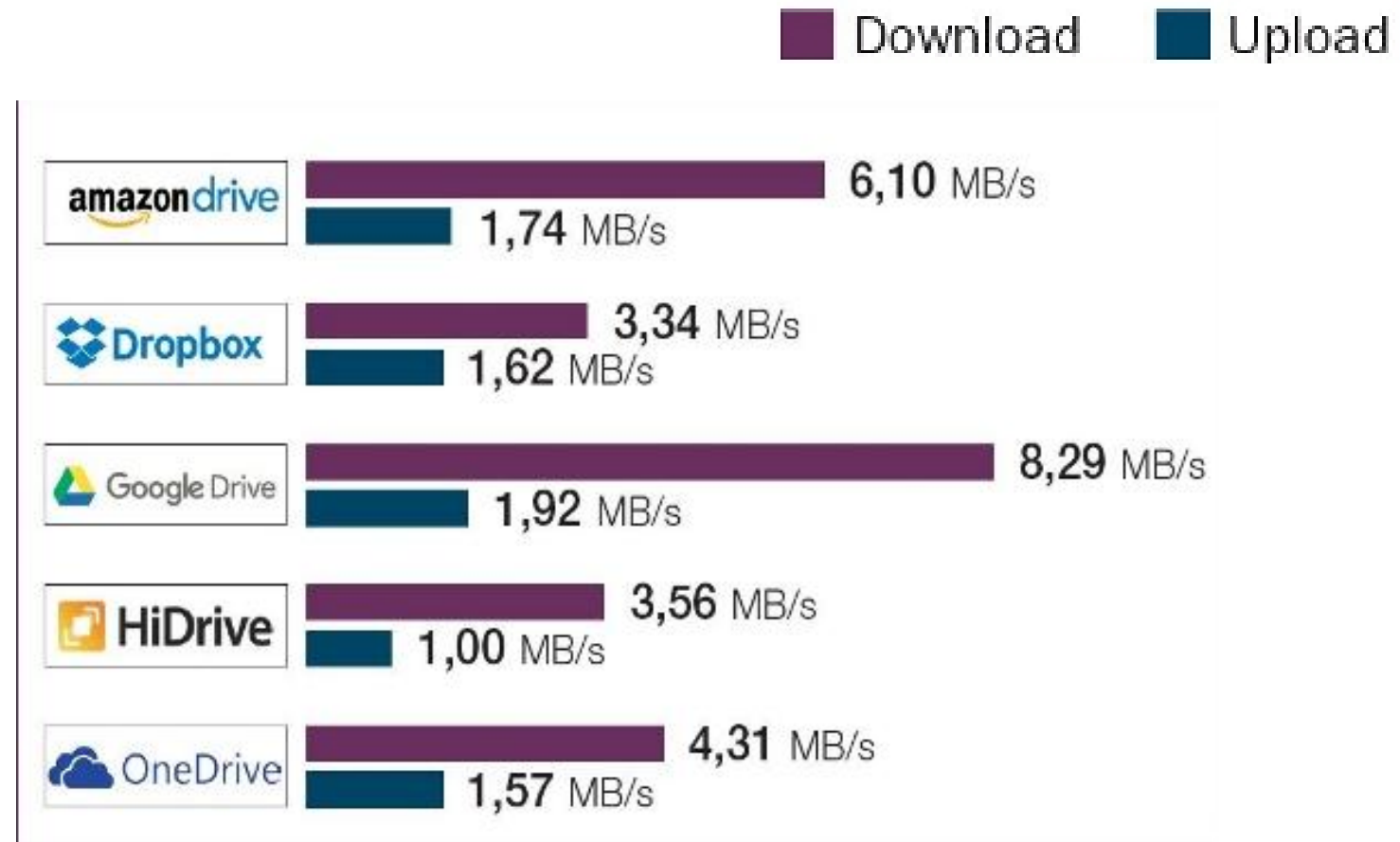
If your target clients are in the same area of your ISP, there is no problem...



If your target customers are in different AS, then the problem may be more complicated because of Internet



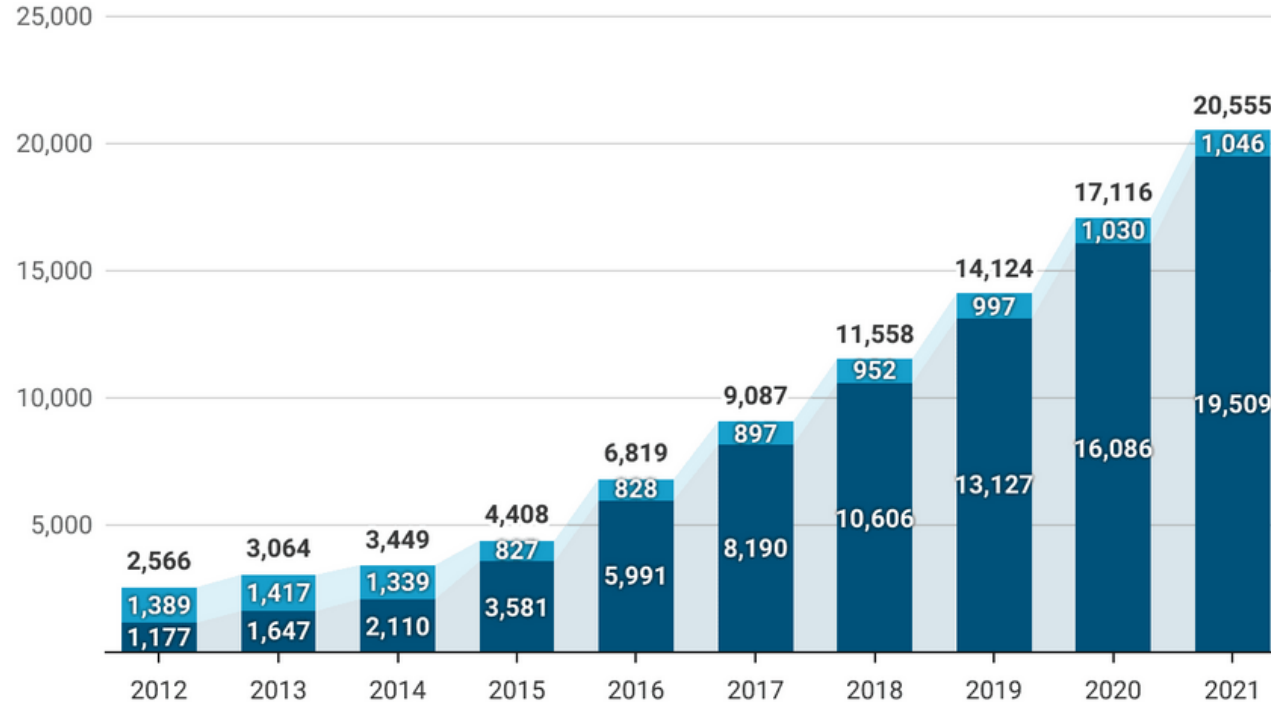
Virtual storage in cloud



Data Center traffic (Annual Growth Rate)

Traffic in exabytes per year

Cloud Data Center Traditional Data Center



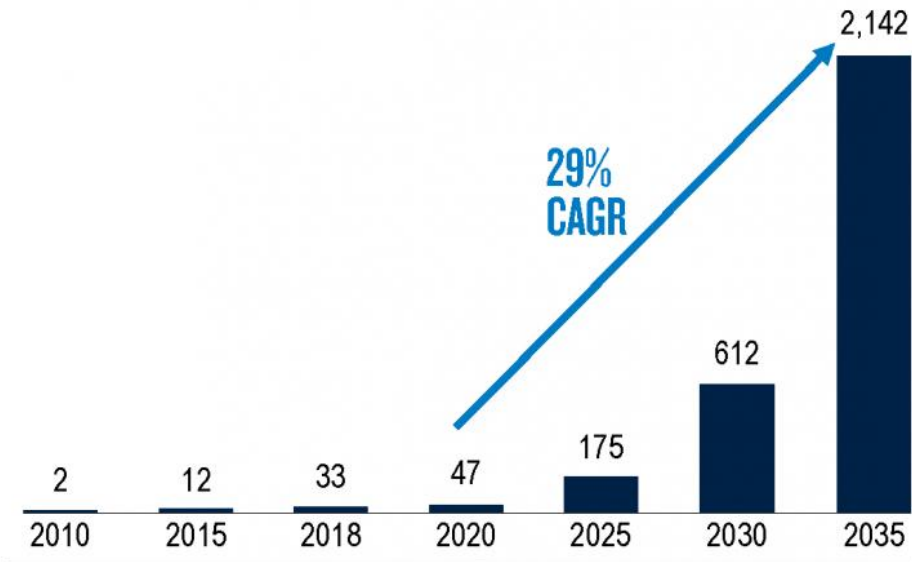
1000 **kilobytes** = 1 Megabyte
1000 **Megabytes** = 1 Gigabyte
1000 **Gigabytes** = 1 Terabyte
1000 **Terabytes** = 1 Petabyte
1000 **Petabytes** = 1 Exabyte
1000 **Exabytes** = 1 Zettabyte
1000 **Zettabytes** = 1 Yottabyte
1000 **Yottabytes** = 1 Bronobyte
1000 **Bronobytes** = 1 Geopbyte



Expected growth of data centers

Exponential growth of data creation

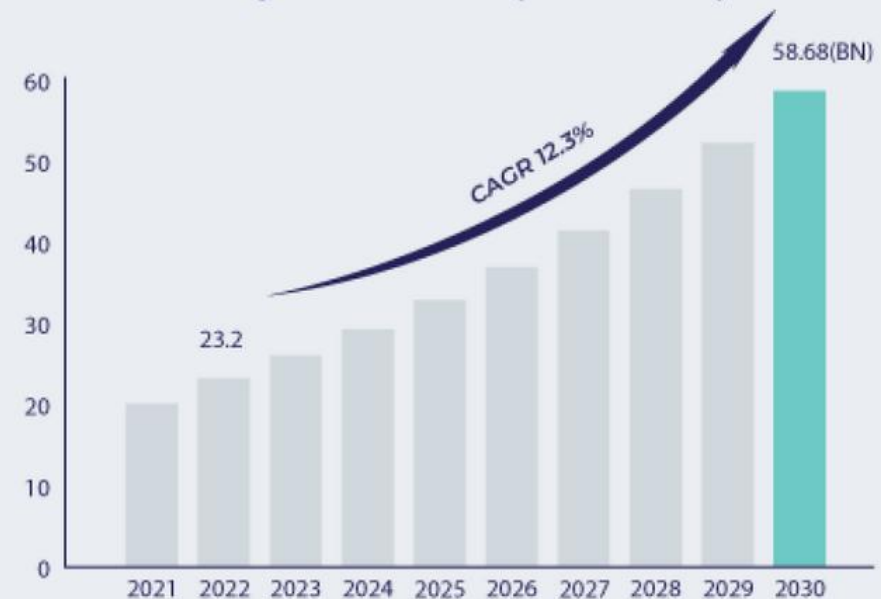
Worldwide data creation in zettabytes¹



$$\text{CAGR Formula} = \left[\left(\frac{\text{Ending Value}}{\text{Beginning Value}} \right)^{\frac{1}{\text{No. of years}}} - 1 \right] \times 100\%$$

Compound Annual Growth Rate

Global Data Center Networking Market Size by 2023 to 2030 (USD Billion)



Effects

As more software goes to the cloud, the client operating system loses relevance because the browser becomes **THE INTERFACE**

With the advent of cloud-based productivity software, the importance of MS Office and other desktop software may decrease

- + There could be the possibility of creating and endorsing open standards that everyone can benefit from
- The market dominance that may come as the result of a positive network effect can also have the negative result of creating a single target for ill-intentioned users. If everyone uses Amazon's cloud storage, then that is the only place where someone needs to break into if they want to steal information. The burden on the provider to maintain security and reliability can increase to the point where adding new users actually decreases the total value



The network impact for the *consumer*

1. Last mile problem

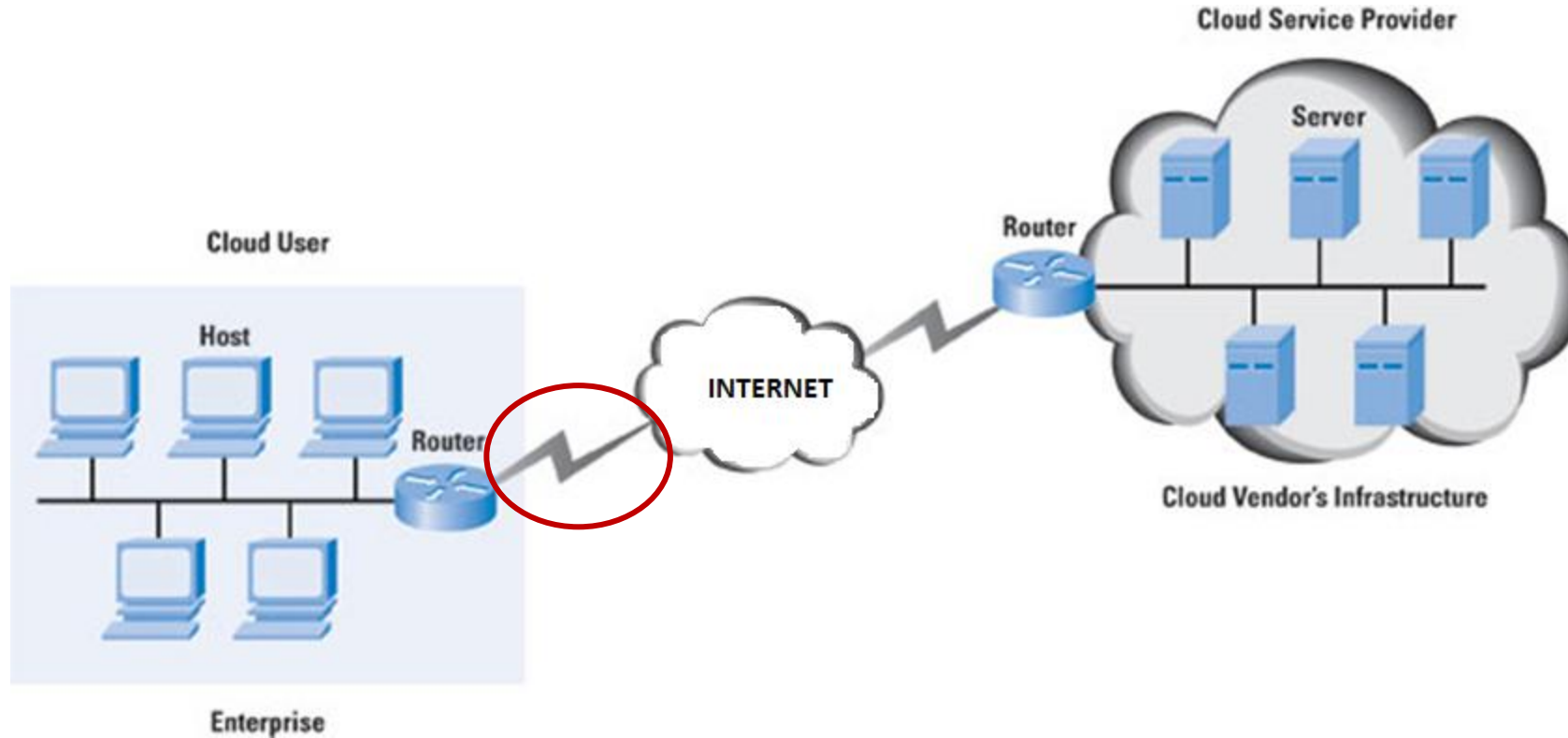
2. Internet traffic

- If you intend to use/integrate external services for your service applications, **bandwidth throughput** to your business locations should be a very significant factor in your decision-making process
 - ➔ If you can choose, then pick the data center that is located in the best connected AS
 - ➔ **If you are not well connected, avoid to put all of your IT services in the cloud** (a good, possible multiple Internet connection is a key parameter to choose a business location for your servers)

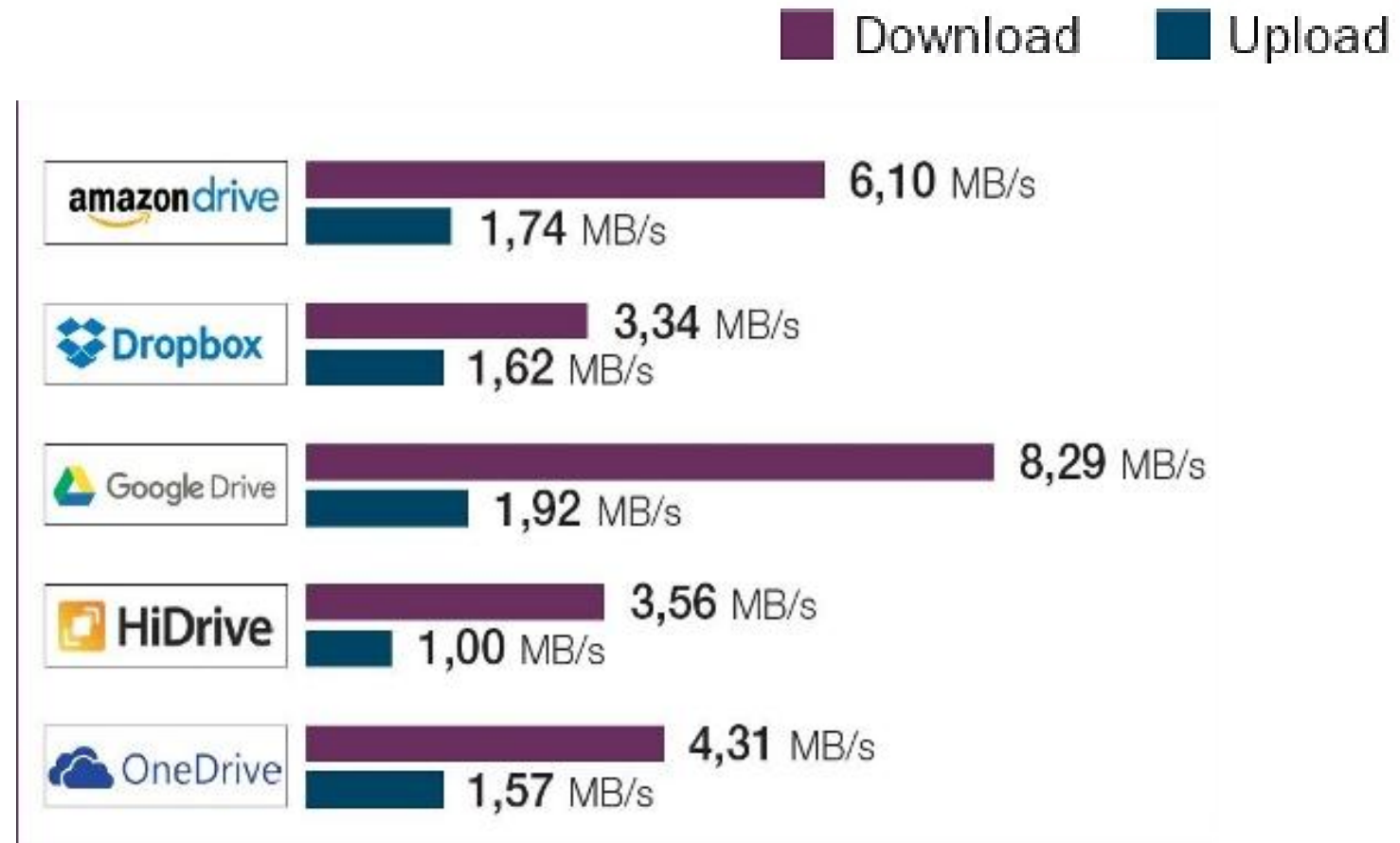


Scenario 1: Cloud for enterprise use

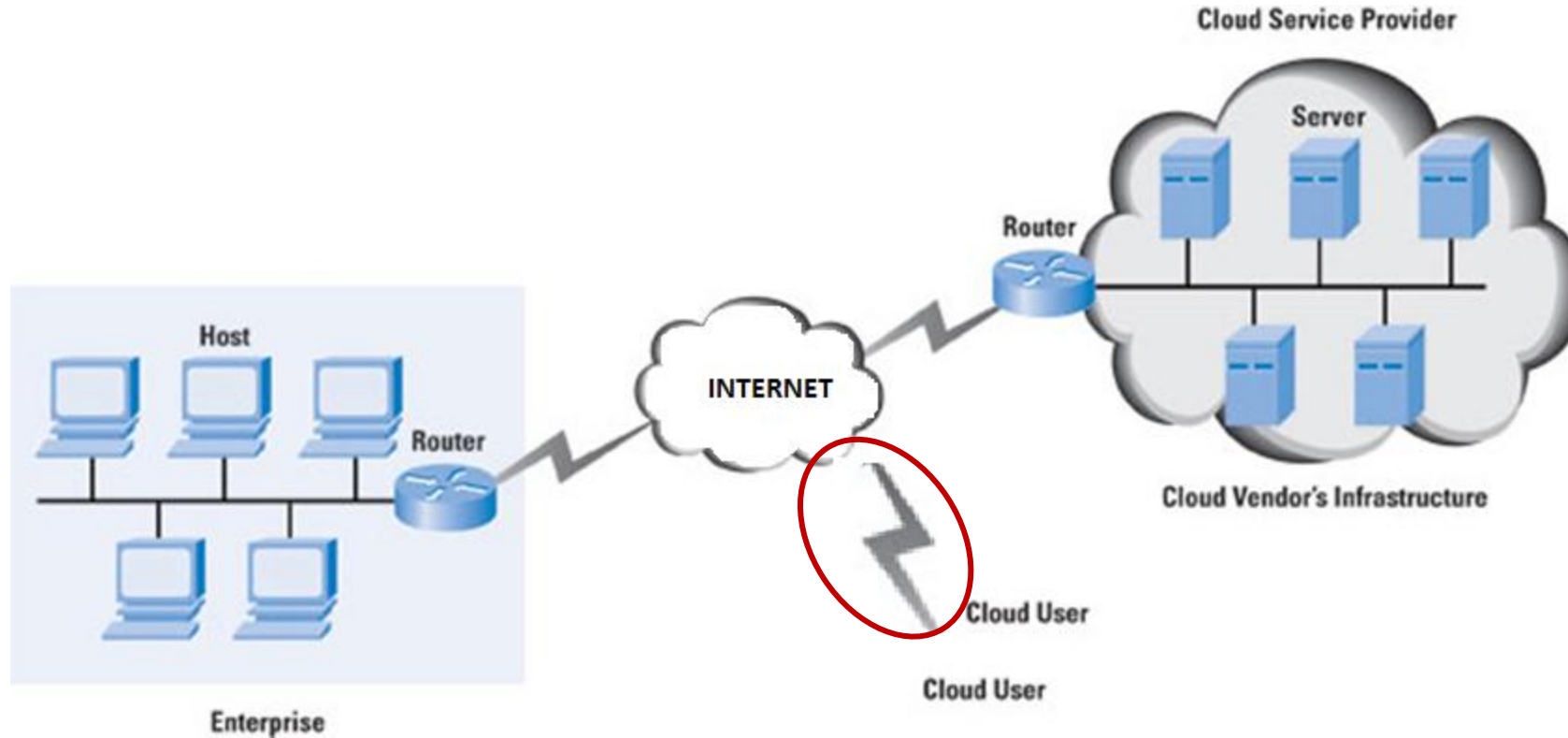
(e.g., email, Office 365, storage, CRM)



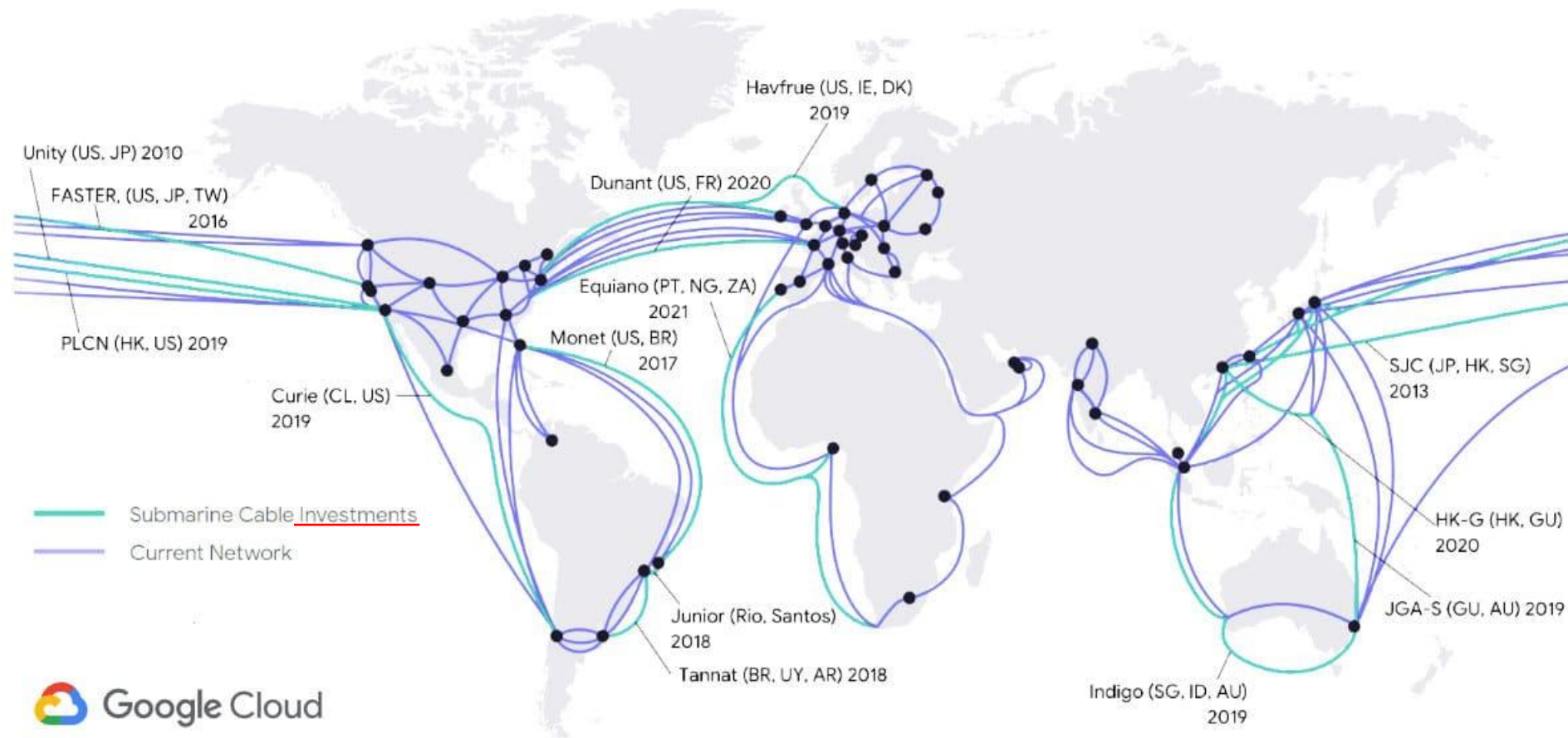
Virtual storage in cloud



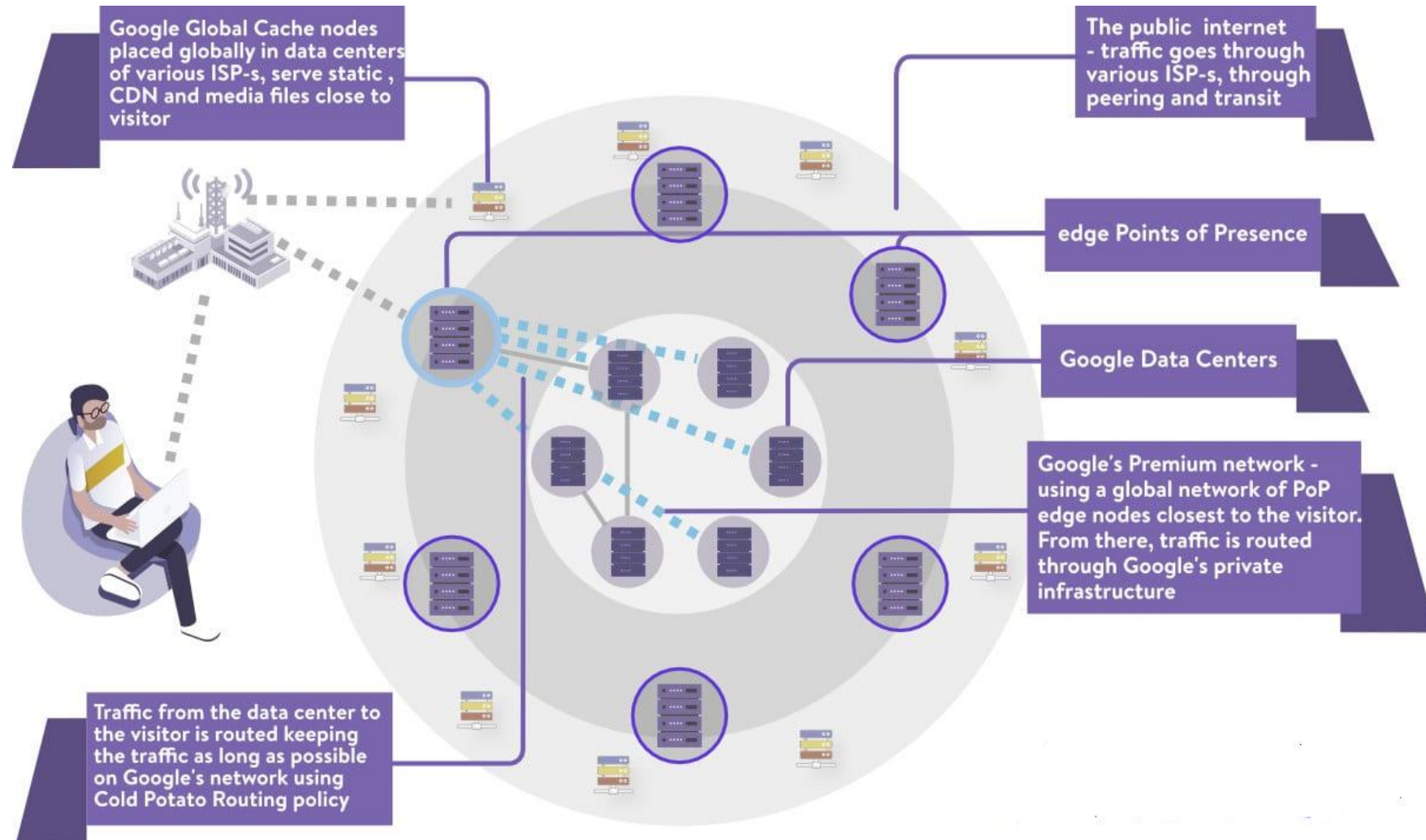
Scenario 2: Cloud for external services



Private WAN networks (*not for everybody*)

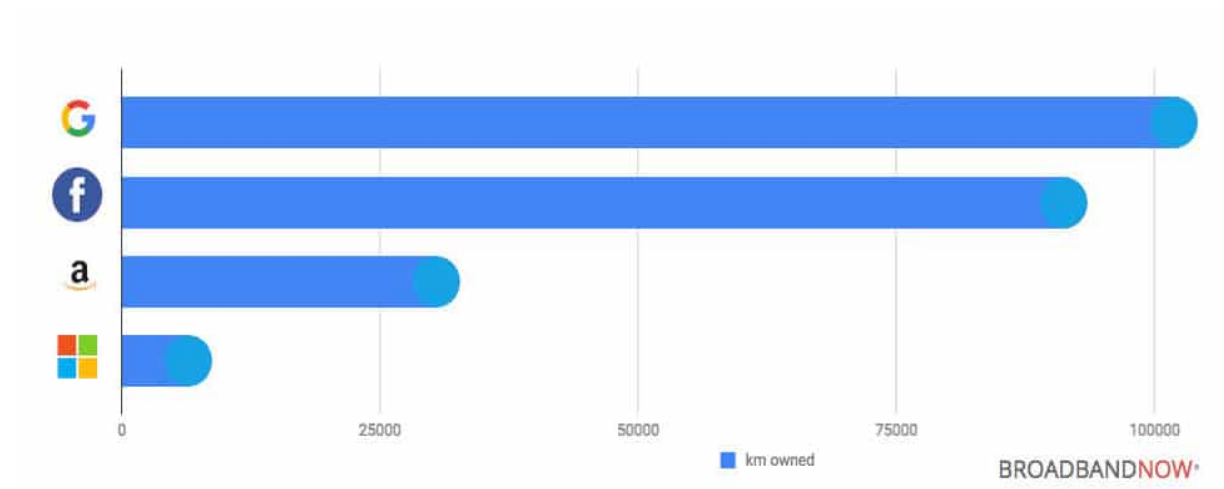


Traffic management in Google



Other cloud providers ...

... are following the Google path



Submarine cables (in Km)
owned by Cloud providers

- Tend to establish partnerships with Tier-1 and Tier-2 ISPs
- Some cloud providers offer private (direct) connections with their ***Points of Presence*** (POP)

The most recent project by Meta: a 50,000 Km cable with a geo-political message (*bring good connections where the future customers will live*)



Cloud services



Inspiration: the historical 4 utilities

[John McCarthy, MIT Centennial, **1961**]: “**Computing may someday be organized as a public utility**”

(1) Water



(2) Electricity



(3) Gas



(4) Phone/Internet



Implementation of the initial vision

[Jeff Bezos, CEO Amazon, **2006**]:

“It’s important to sell unused resources of a datacenter so a cost may become a profit”

→ This represents the birth of CLOUD COMPUTING

[Steve Ballmer, CEO Microsoft, **2016**]:

“The cloud is the future, everything else is accessory”



Two main paradigms

Service

Deployment

Computer science and engineering are not commodities



Service paradigms (3 *prevalent* schemes)

- ***Infrastructure as a Service*** (IaaS)
- ***Platform as a Service*** (PaaS)
- ***Software as a Service*** (SaaS)



Amazon (AWS): IaaS + PaaS

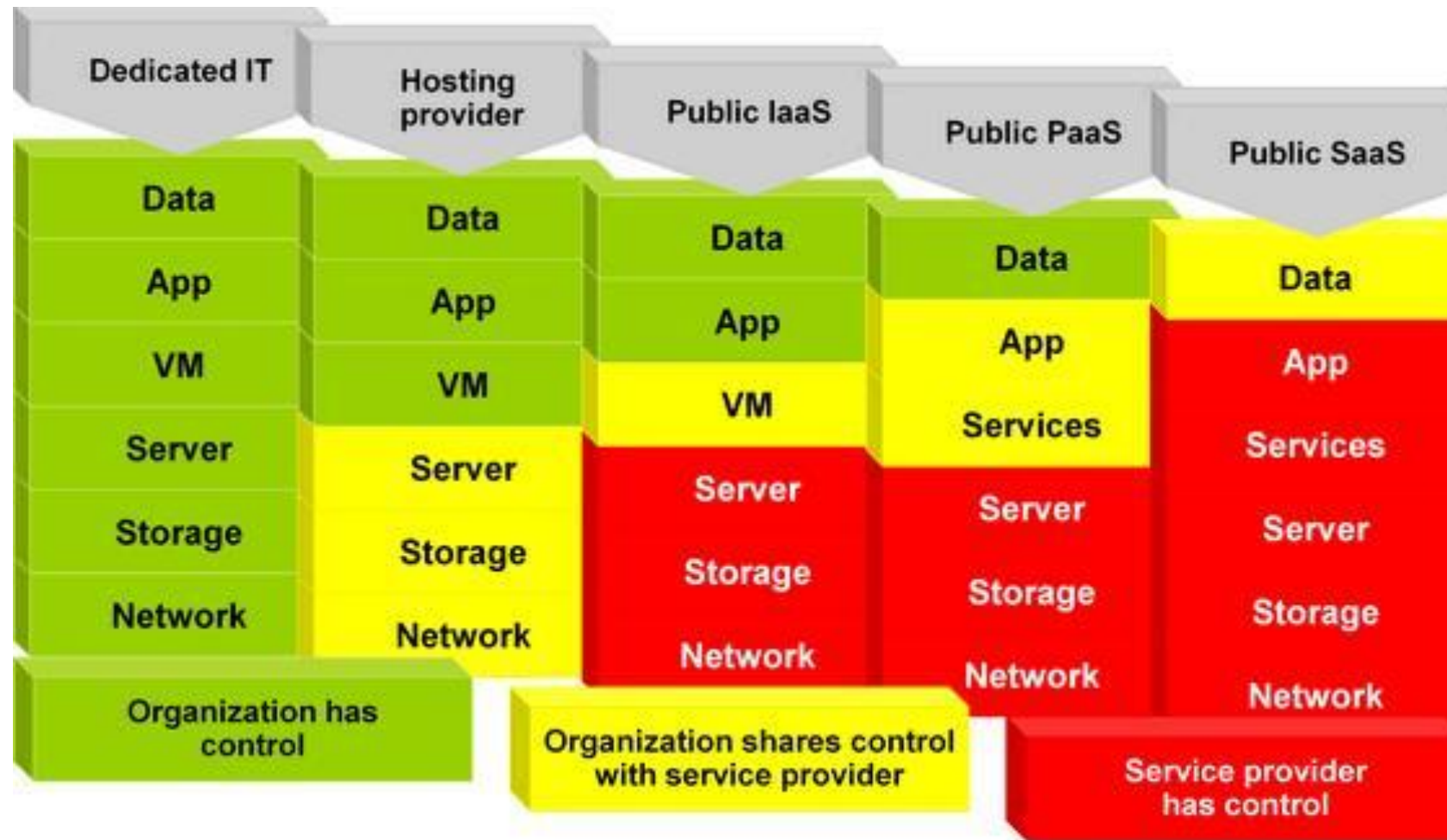
E-Commerce Service (ECS)	Alexa Web Informat. Service (AWIS)	Alexa Web Services Platform (AWSP)	Mechanical Turk (MT)	Historical Pricing (AHP)	Simple Queue Service (SQS)	Alexa Top Sites	Amazon Web Sites Amazon.com Amazon.jp Amazon.fr Amazon.ca Amazon.de Amazon.co.uk joyo.com
AWS – Amazon Web Services Shared Code							
Amazon Technology Platform							

A huge step ahead for Amazon's engineers. This is much more than offering housing or hosing services. **This means sharing the same resources and services of you with your customers**

Selling services. Which ones?

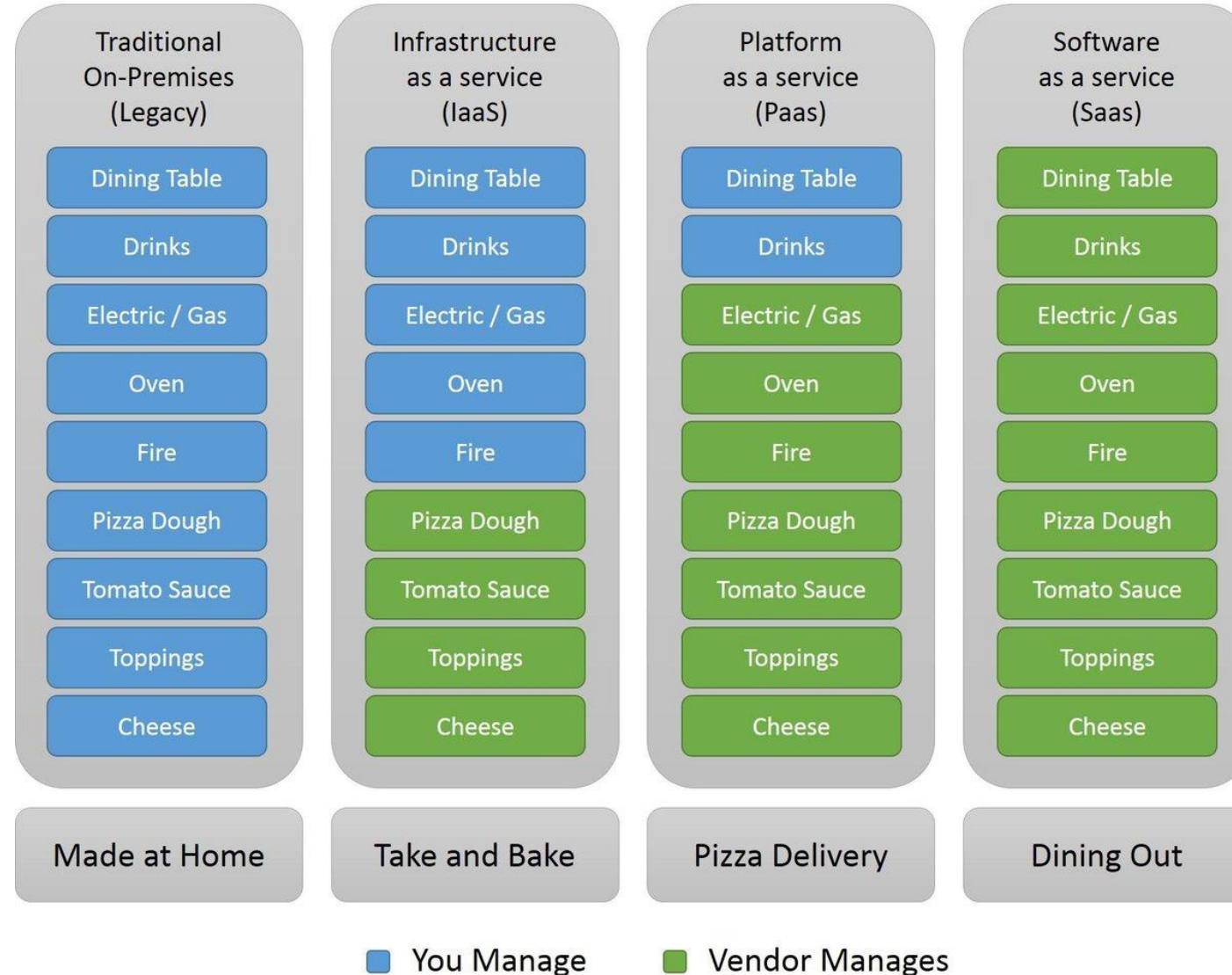


Cloud services: *which organization is responsible of what?*



Metaphor in a quite different context

Pizza as a Service



Deployment models (2 *prevalent*)

Public cloud

The cloud infrastructure is made available to the general public or a large industry group and is owned by an organization (i.e., **cloud provider**) selling one or more types cloud services

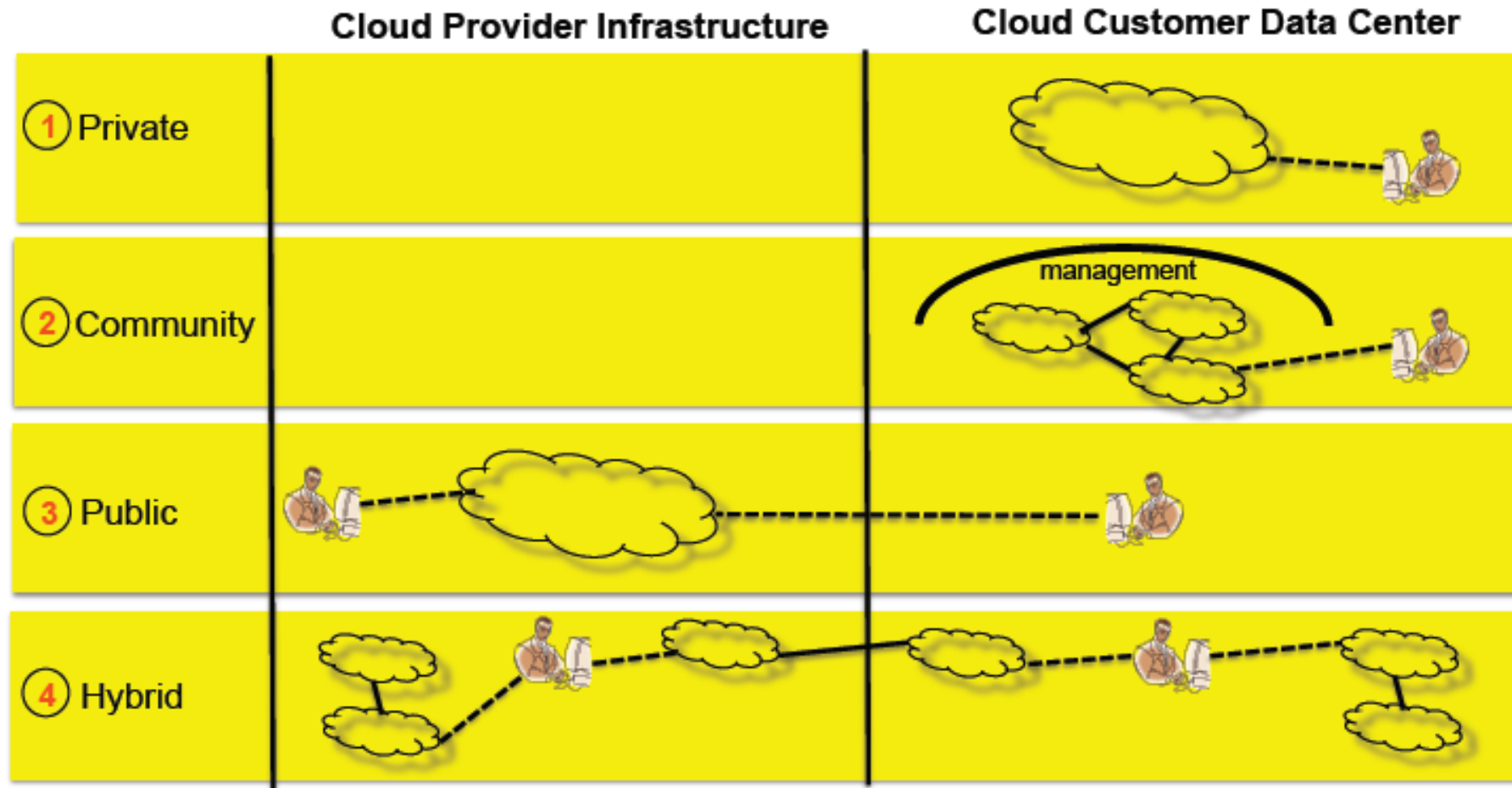
Private cloud

The cloud infrastructure is operated for one organization. It may be managed by:

- the organization itself (*private cloud*)
- or a third party (*exclusive cloud*) and may exist on-premise or off-premise



Deployment models



Deployment models

1. Public cloud

2. **Community cloud**. The cloud infrastructure is shared by several organizations and supports a specific community that has shared concerns (e.g., business mission, security requirements, policy, and compliance considerations). It may be managed by the organizations or by a third party and may exist on premise or off premise

3. Private cloud

4. **Hybrid cloud**. The cloud infrastructure is a composition of two or more clouds (private, community, or public) that remain unique entities but are bound together by standardized or proprietary technology that enables data and application portability

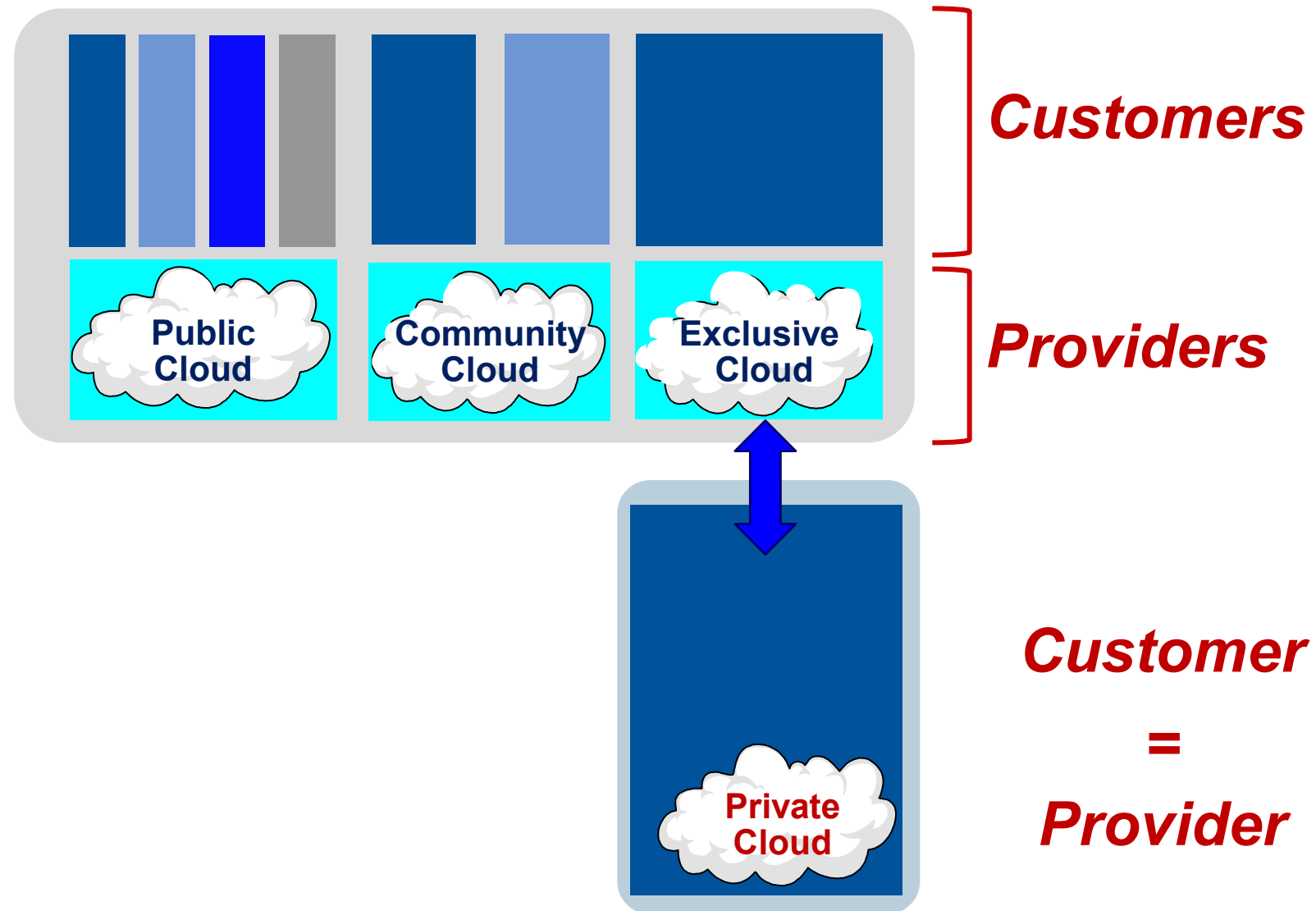


Community clouds – *many possible examples*

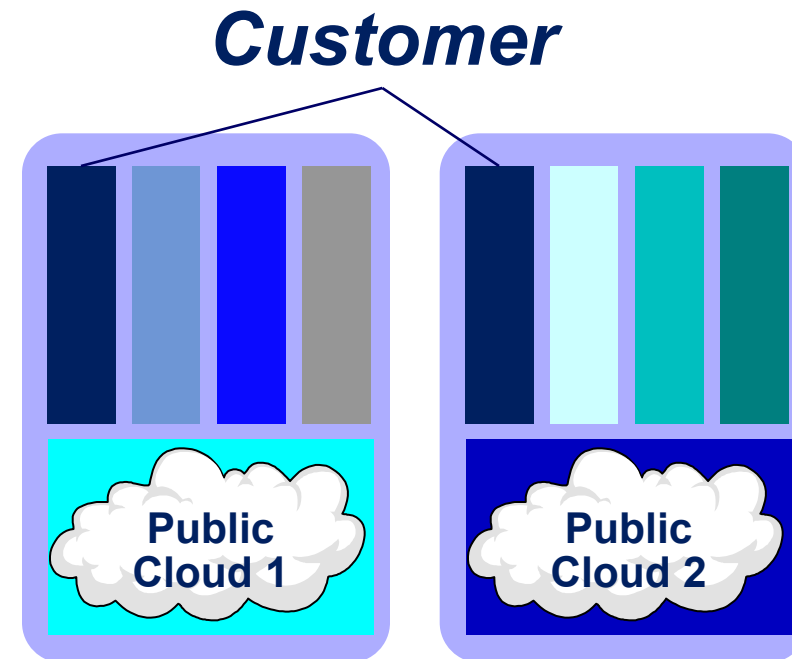
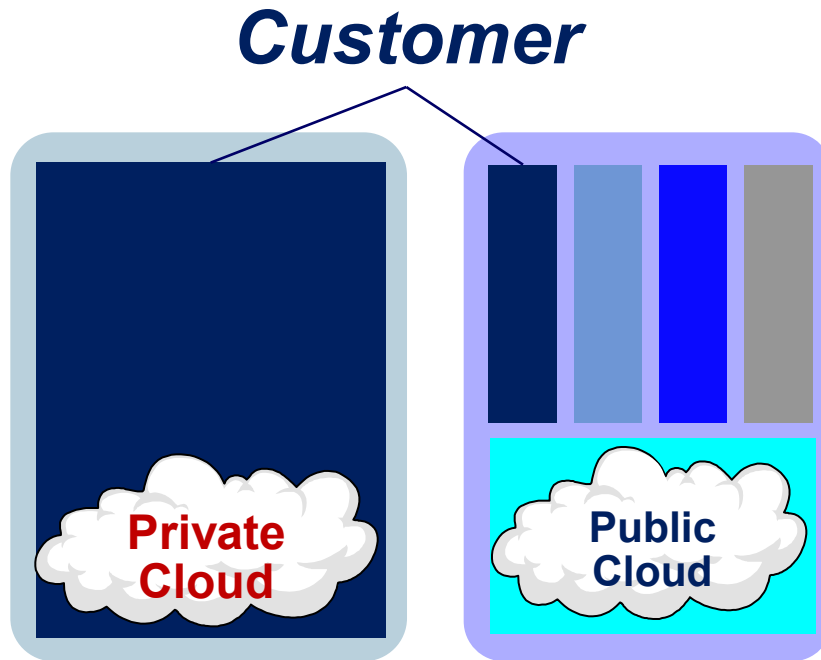
- Public Administration
- Health care
- Small-Medium Enterprises belonging to the same industrial sector
 - Vertical industries
 - Established Providers-Clients
- ...



Deployment models

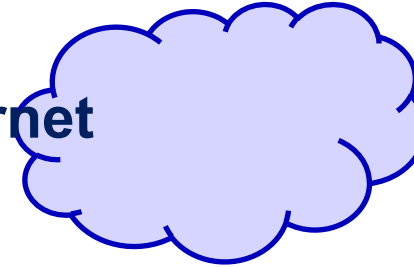


Hybrid solutions



The term *Cloud computing*

Cloud = is also a symbol of Internet



computing = *Everything as a Service*
(EaaS o XaaS)

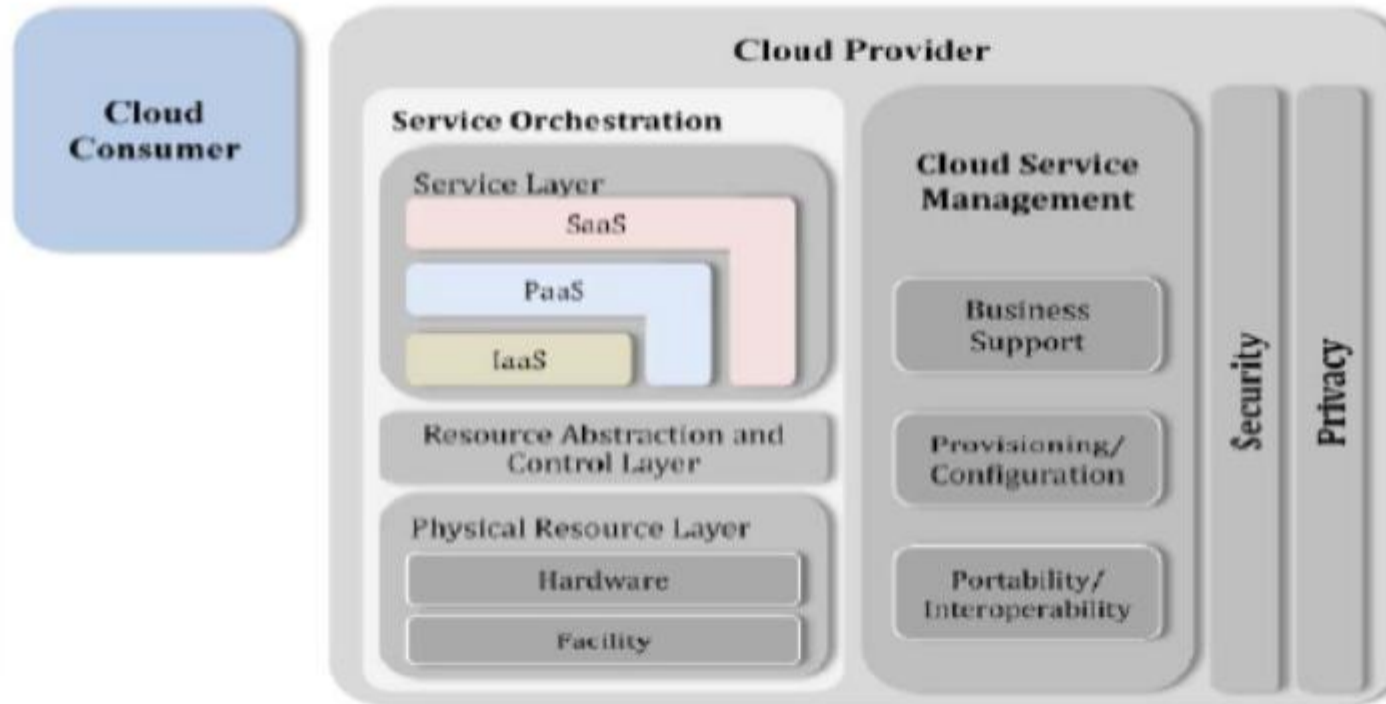


Offered vision of the cloud

1. **A logically isolated set of resources** (disk, memory, processor, network)
2. **Elastic** (*scalability and reverse scalability*)
3. **Reliable** (*always available resources*)
4. **Secure**
5. **Pay-per-use cost model**



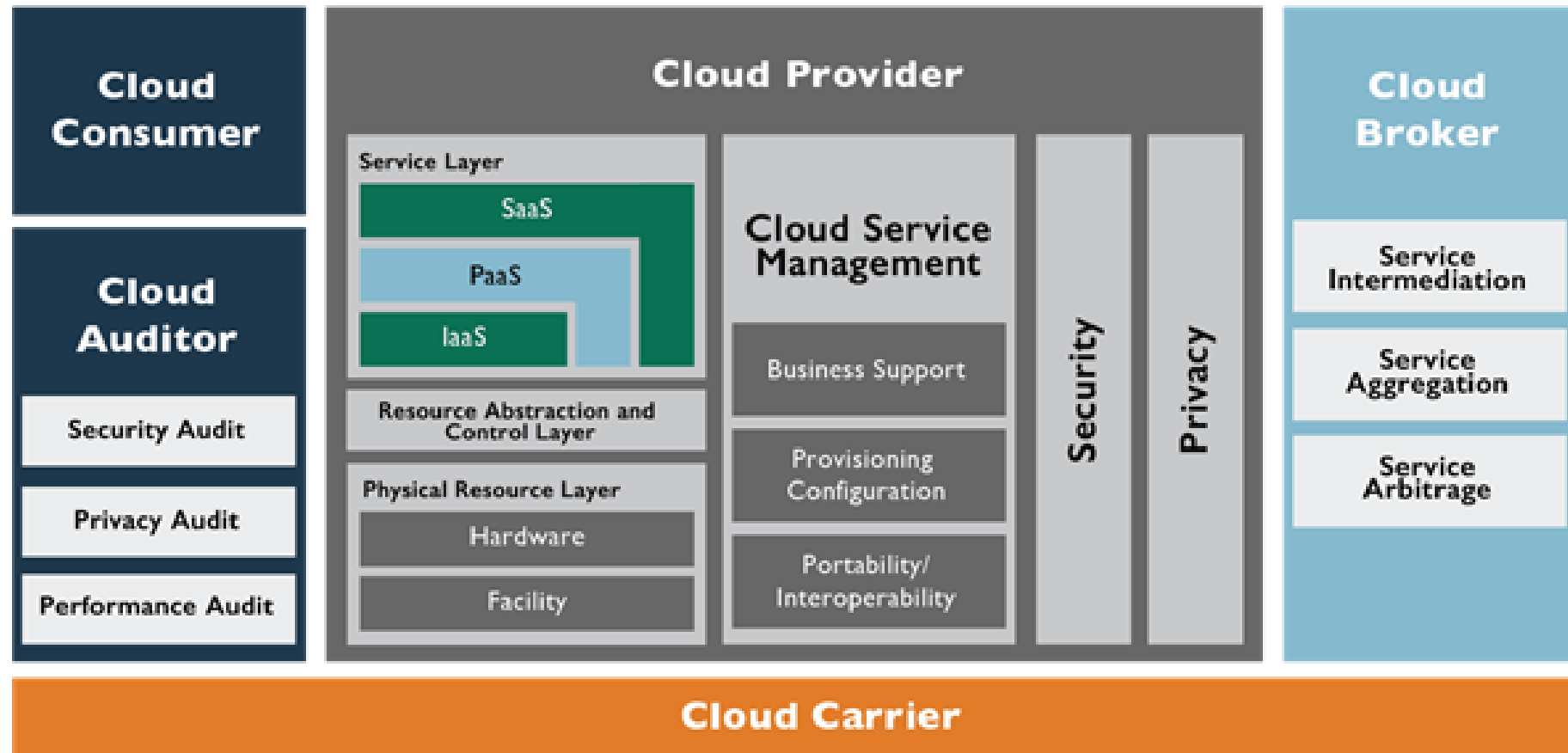
NIST definitions



“Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a **shared pool of configurable computing resources** (e.g., networks, servers, storage, applications, and services) that can be **rapidly provisioned and released** with minimal management effort or service provider interaction. This Cloud model promotes **availability, elasticity, security** [...].”



NIST definitions: overall picture



The third alternative to *Make* or *Buy*



Cloud providers



Cloud provider companies

Only a handful of major IT players can really build and manage an infrastructure of multiple geographically distributed IDCs

amazon.com



Microsoft

Windows Azure Platform



Google Cloud Platform



Worldwide telco operators are trying to access the cloud (IaaS) market, but



Today, other big IT companies are reaching the pack

ORACLE®

Cloud providers specialized in *content delivery*

Akamai → Content delivery (market leader) and security services



Cloudflare → Security (35% DNS, API and SaaS protection, DDoS protection, SSL communications) and Content delivery through *anycast* technology



The big players [Gartner, 2013 and 2014]



The distance augments [Gartner, 2016]

(just MS and Google improve, but they remain behind AWS)

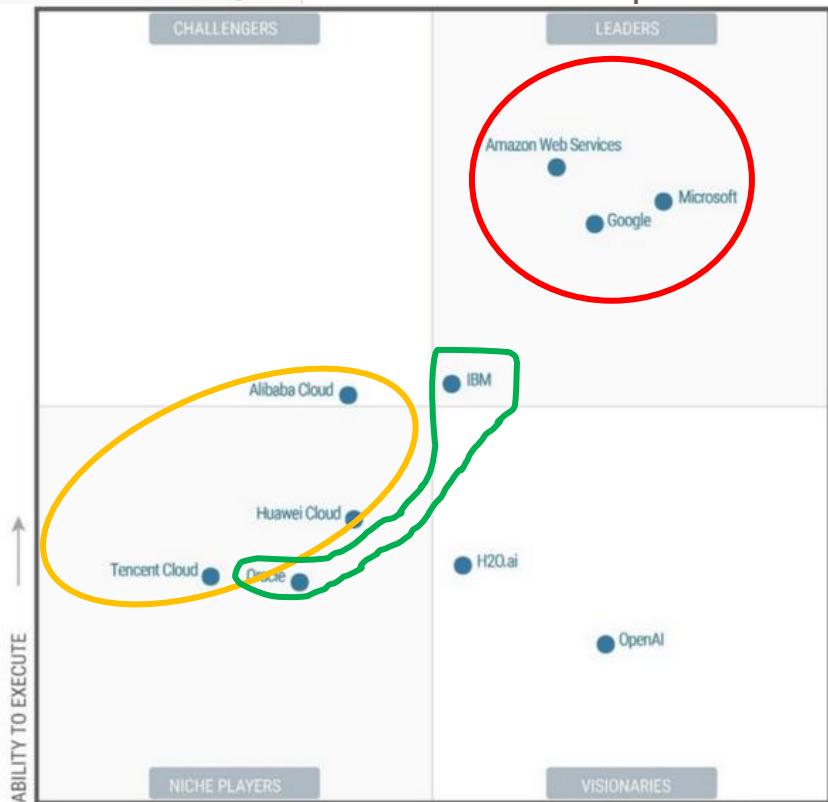
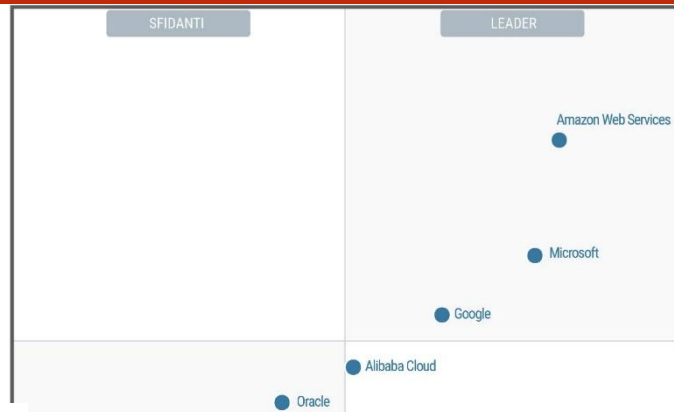
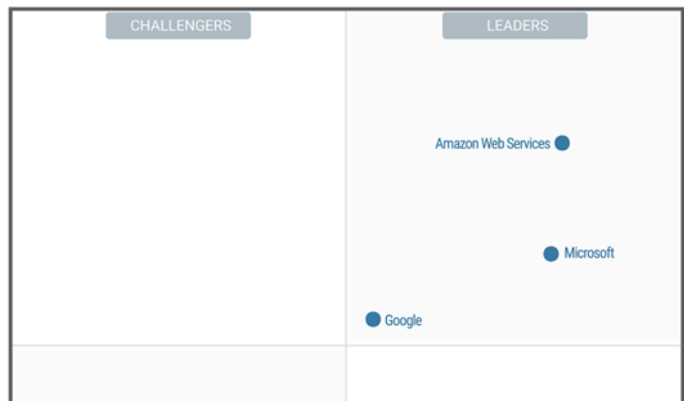
Figure 1. Magic Quadrant for Cloud Infrastructure as a Service, Worldwide



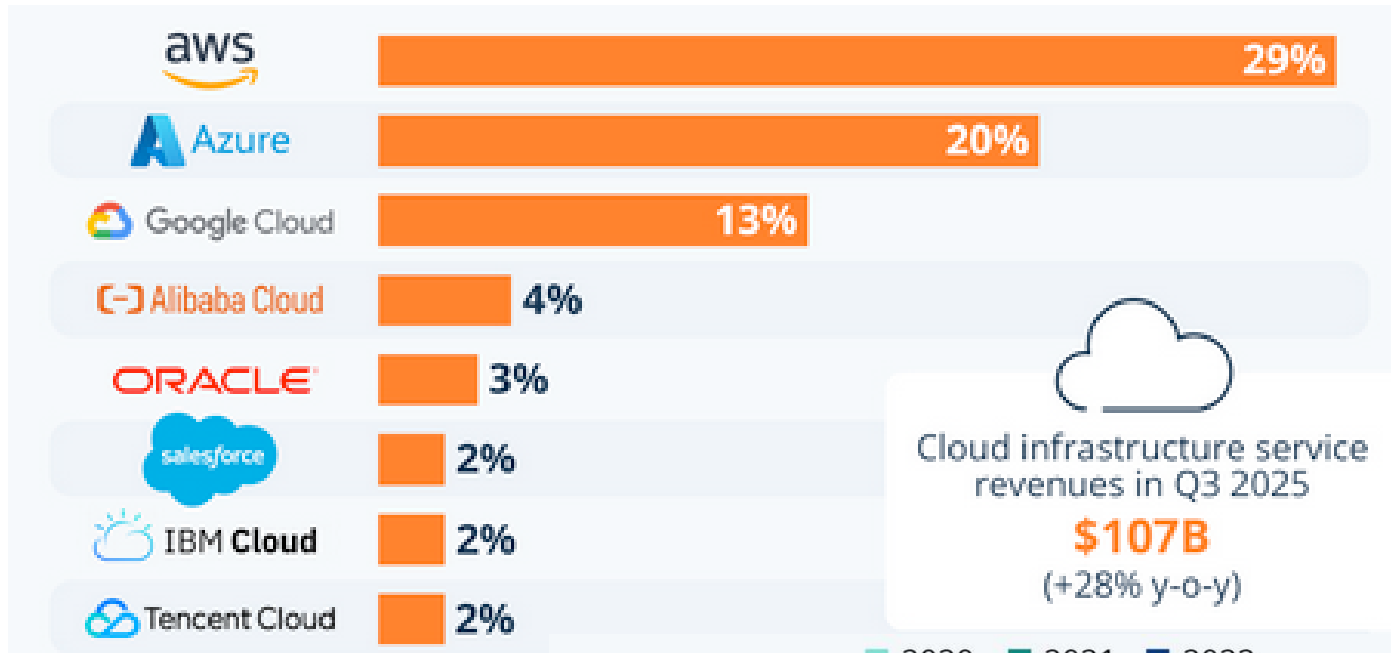
Source: Gartner (August 2016)



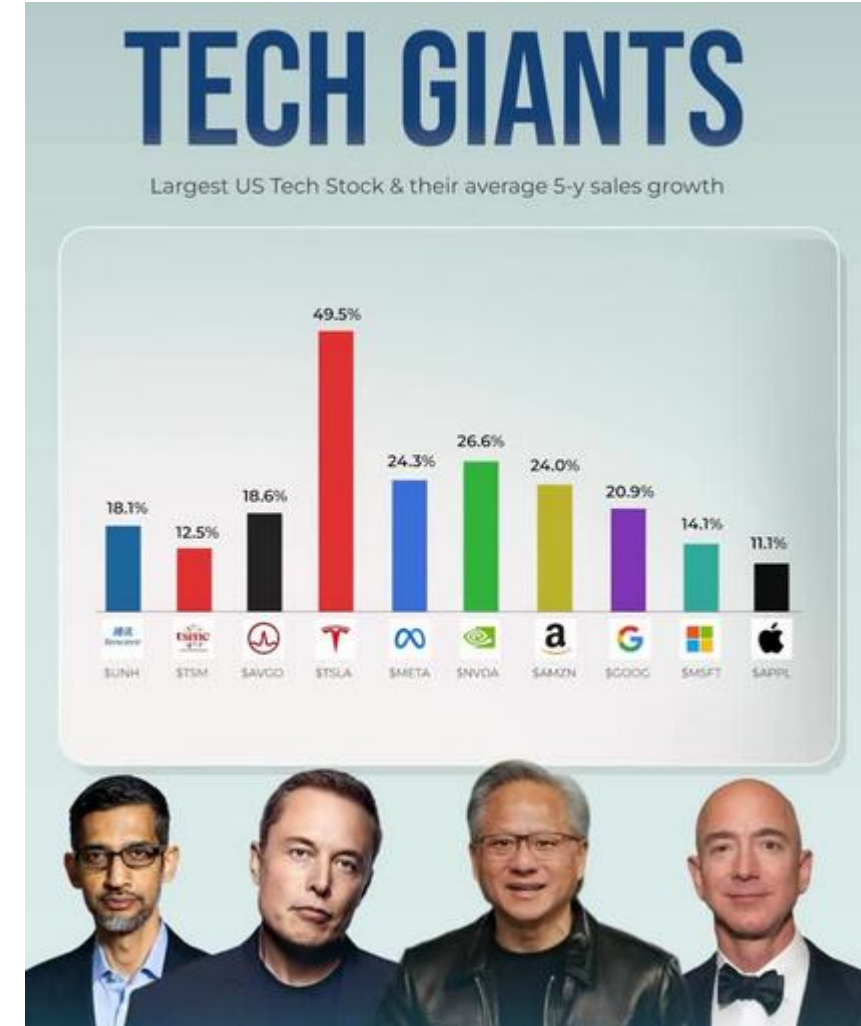
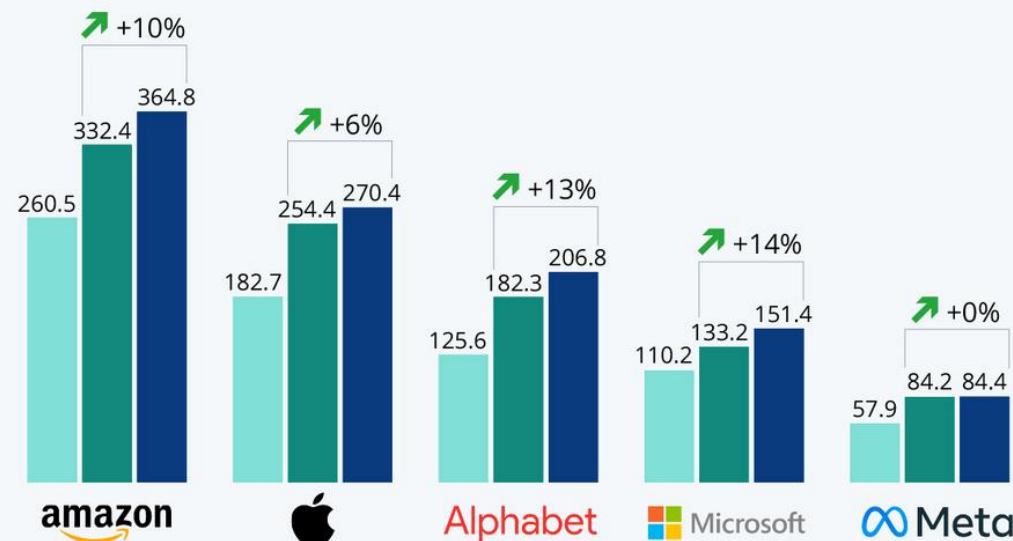
The top players [Gartner, 2019-2024]



Continuous growth



* Includes platform as a service (PaaS) as well as hosted private cloud services



Cloud clear benefits

- **Scalability**
- **Reliability**

Cloud clear disadvantages

- **Vendor lock-in** → *Unavoidable*
- **Cost** → *FinOps*

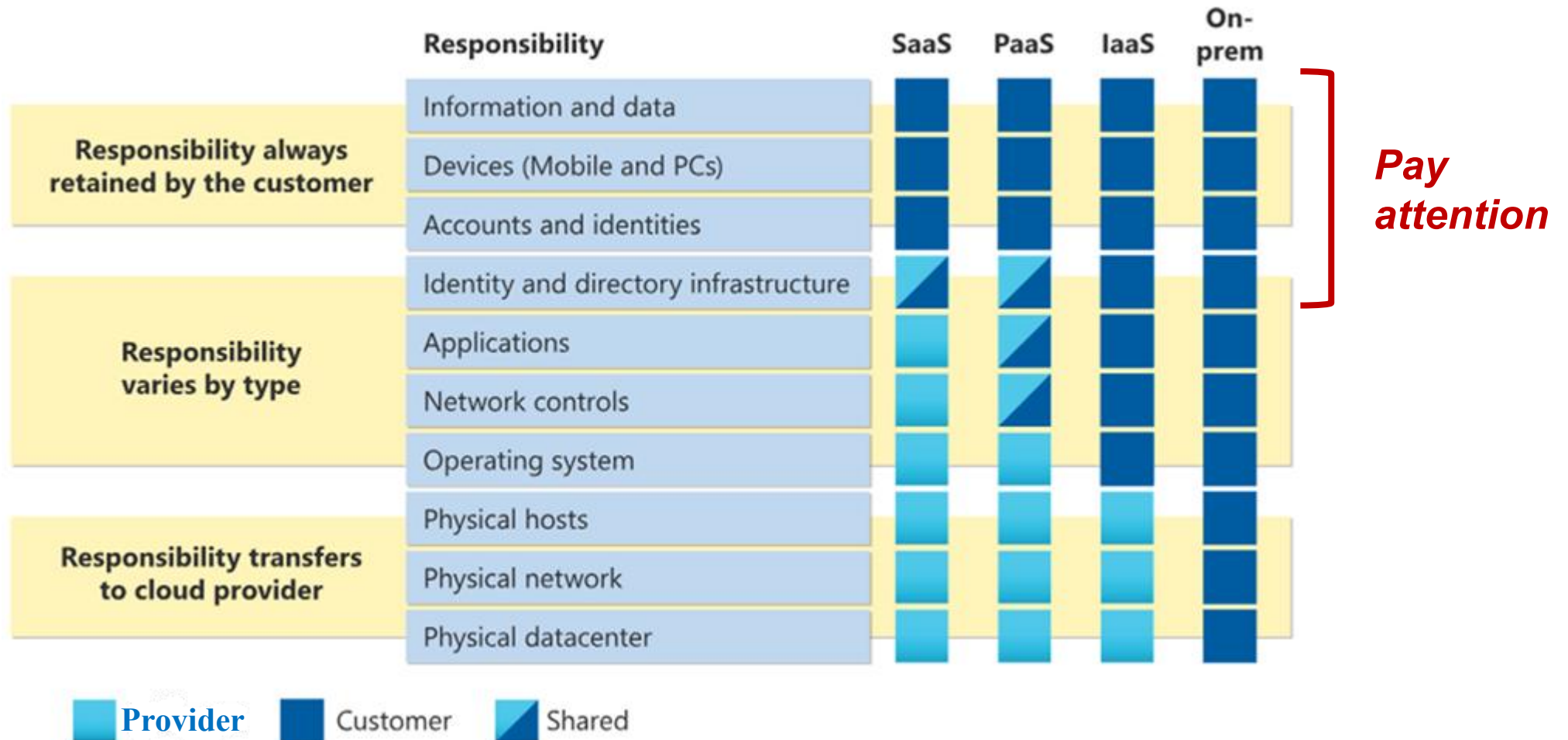


Cloud: A shared responsibility



Responsibility

In an on-premises datacenter, you own and have to manage the whole stack.
As you move to the cloud, some responsibilities (not all) transfer to your provider



Example Azure

Responsibility area	On-premises	IaaS	PaaS	SaaS
Customer data	Customer	Customer	Customer	Customer
Configurations and settings	Customer	Customer	Customer	Customer
Identities and users	Customer	Customer	Customer	Customer
Client devices	Customer	Customer	Customer	Shared
Applications	Customer	Customer	Shared	Shared
Network controls	Customer	Customer	Shared	Microsoft
Operating system	Customer	Customer	Microsoft	Microsoft
Physical hosts	Customer	Microsoft	Microsoft	Microsoft
Physical network	Customer	Microsoft	Microsoft	Microsoft
Physical datacenter	Customer	Microsoft	Microsoft	Microsoft



Your (*customer*) responsibilities

Regardless of the type of deployment, you always retain the following responsibilities:

- **Data** – You are responsible for your data, including data classification, data protection, encryption decisions, and compliance with data governance requirements
- **Endpoints** – You are responsible for protecting client devices and endpoints that access your cloud services, including mobile devices, laptops, and desktops
- **Accounts** – You are responsible for managing user accounts, including creating, managing, and removing user access (*don't forget*)
- **Access management** – You are responsible for implementing and managing access controls, including role-based access control (RBAC), multifactor authentication, and conditional access policies



Shared responsibilities

Some responsibilities are shared between you and the provider, with the division varying by service model:

- **Applications** - In IaaS, you are fully responsible for deployed applications. In PaaS and SaaS, the provider manages parts of the application stack, but you are responsible for application configuration, code security, and access controls
- **Network controls** - In IaaS, you configure all network security including firewalls and network segmentation. In PaaS, the provider typically provides baseline network security, but you configure application-level network controls. In SaaS, the provider manages internal network security.
- **Client devices** - In SaaS scenarios, Google, Microsoft and AWS may provide some device management capabilities, but you are responsible for endpoint protection and compliance



Provider responsibilities

- **Physical security** - Securing datacenters, including facilities, physical access controls, and environmental controls
- **Physical network** - Managing network infrastructure, including routers, switches, and cables within datacenters
- **Physical servers** - Managing and maintaining the physical servers that host cloud services
- **Hypervisor** - Managing the virtualization layer that enables virtual machines in IaaS and PaaS
- **Platform services** - In PaaS and SaaS, the cloud provider manages operating systems, runtime environments, middleware and all their continuous updates



AI shared responsibility

When using AI services, the shared responsibility model introduces unique considerations beyond traditional IaaS, PaaS, and SaaS

- The **provider** is responsible for securing the AI infrastructure, model hosting, and platform-level safeguards
- **Customers** remain accountable for how AI is applied within their environment including:
 - protecting sensitive data
 - managing prompt security
 - mitigating prompt injection risks
 - ensuring compliance with organizational and regulatory requirements



Possible advantages (1/2)

The cloud offers significant advantages for solving long standing **information security challenges**. In an on-premises environment, organizations likely have unmet responsibilities and limited resources available to invest in security, which creates an environment where attackers can exploit vulnerabilities at all layers. Common examples of unmet responsibilities in traditional on-premises environments include:

- 1) **Delayed patching** - Security updates are not applied promptly due to limited IT staff or concerns about system downtime, leaving known vulnerabilities exposed
- 2) **Inadequate physical security** - Server rooms may lack proper access controls, environmental monitoring, or surveillance due to budget constraints
- 3) **Incomplete network monitoring** - Organizations may not have tools or expertise to detect intrusions, monitor traffic anomalies, or respond to threats in real time
- 4) **Outdated hardware** - Aging infrastructure may no longer receive security updates from vendors, creating permanent security gaps
- 5) **Insufficient backup and no disaster recovery** - Backups may be infrequent, untested, or stored on-site, leaving data vulnerable to ransomware or physical disasters



Possible advantages (2/2)

In an on-premise scheme, **many security responsibilities are unmet due to limited resources.**

In the cloud-enabled scheme, you can shift day to day security responsibilities to your cloud provider and reallocate your resources. You are also able to apply cloud-based security capabilities for more effectiveness and use cloud intelligence to improve your threat detection and response time. By shifting responsibilities to the cloud provider, organizations can get more security coverage, which enables them to reallocate security resources and budget to other business priorities.

